



Código Binário e a Linguagem dos Computadores

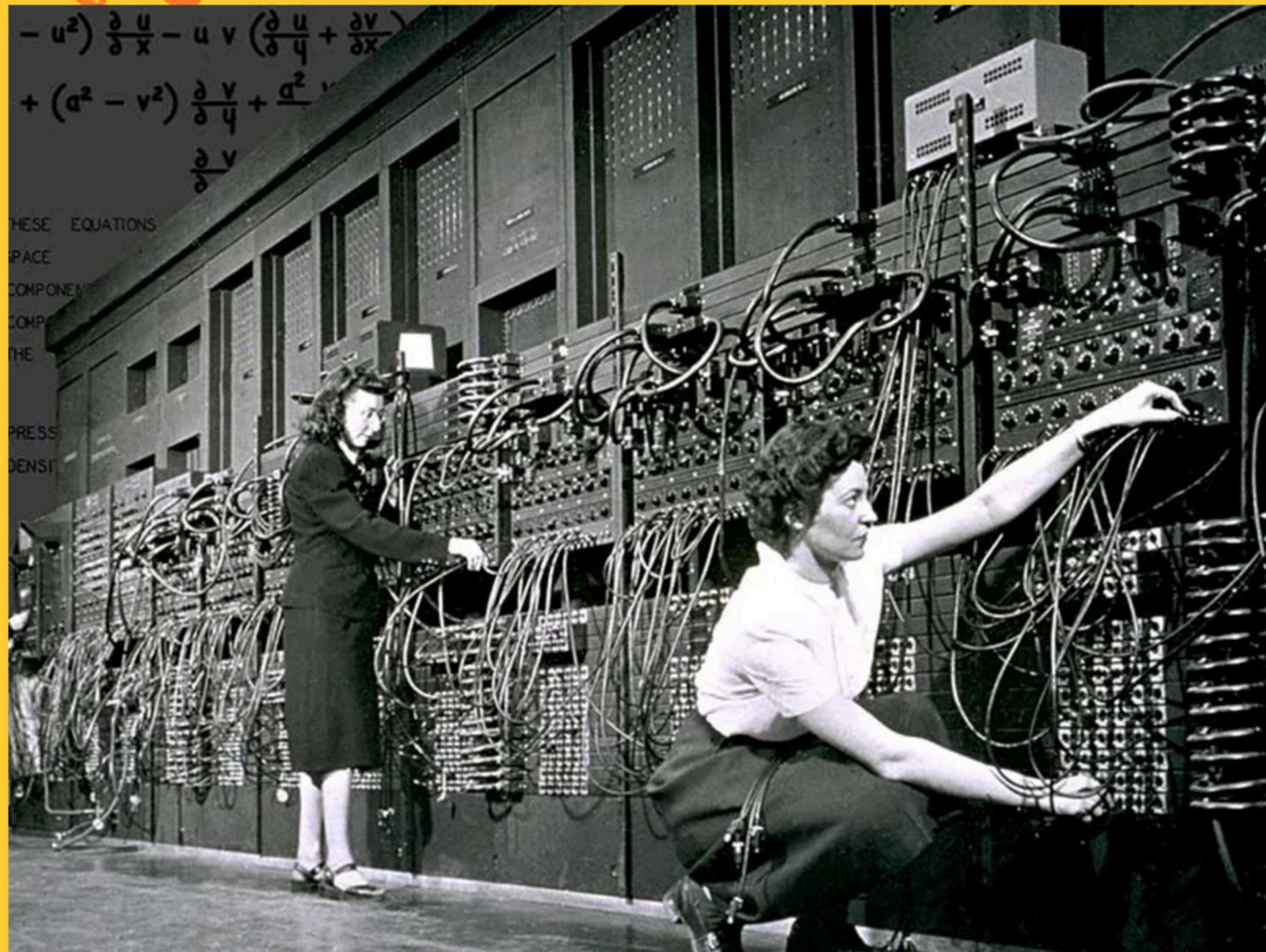




**VOCÊS SABIAM QUE O
PRIMEIRO COMPUTADOR
DIGITAL FOI CRIADO À
APENAS 80 ANOS ATRÁS?**

OLHA ELE AQUI

ENIAC (1946)



A EVOLUÇÃO DOS COMPUTADORES



1ª Geração

2ª Geração

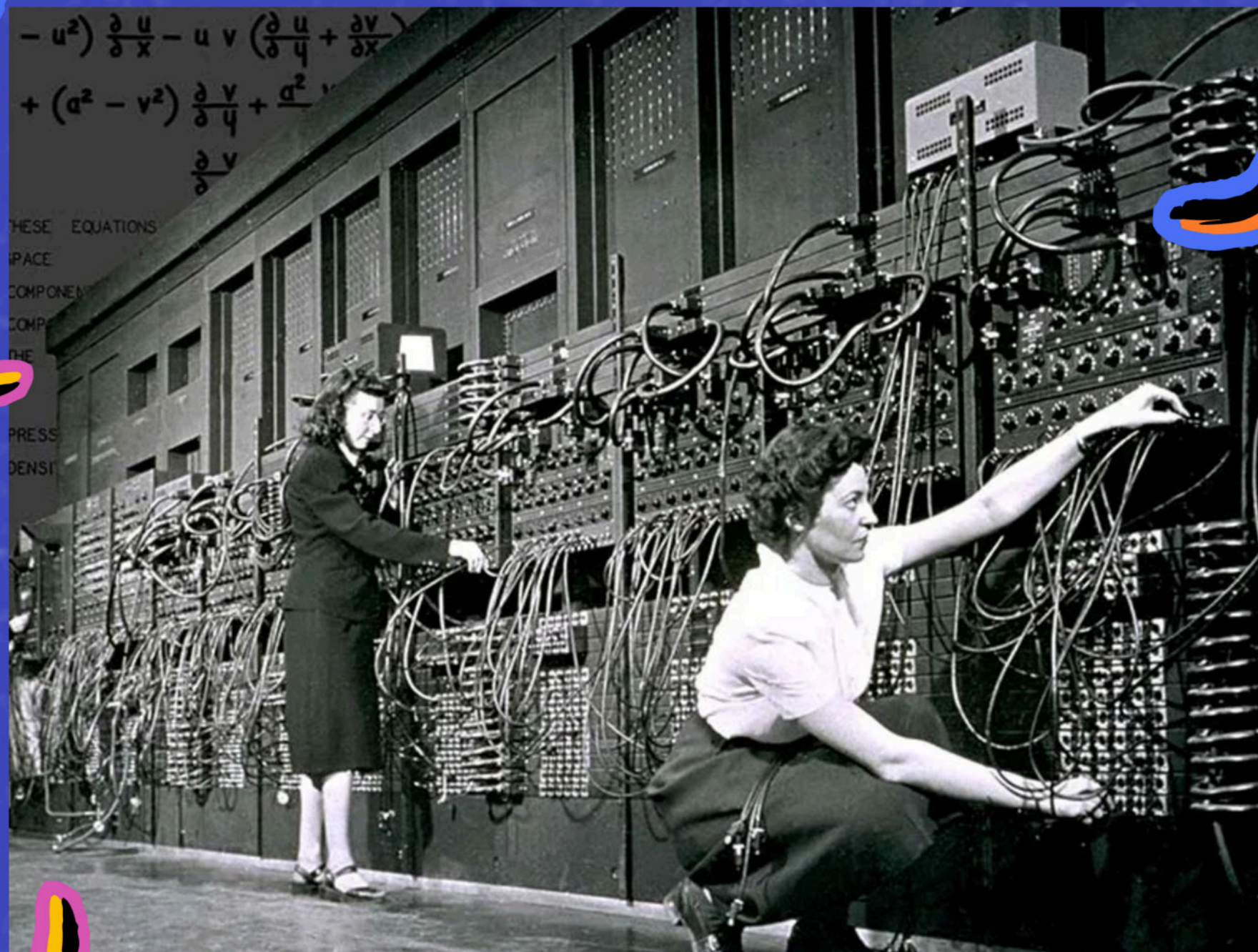
3ª Geração

4ª Geração



Eram imensos,
consumiam muita
energia, eram lentos e
as válvulas quebravam.

Válvulas termiônicas

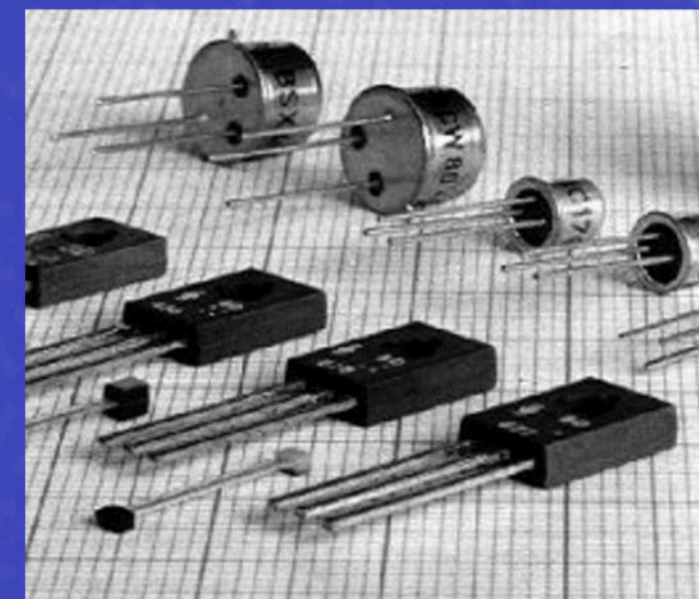


Eniac

1ª Geração

Computadores menores, com processamento mais rápido e muito menos aquecimento.

Uso de transistores

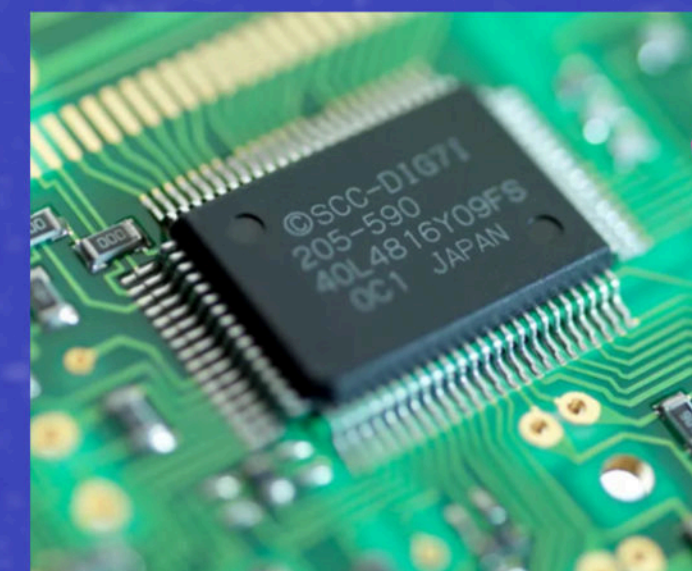


univac
1108

2ª Geração

Computadores muito menores, mais leves, mais baratos, com maior eficiência energética e maior poder de processamento.

Circuito Integrado

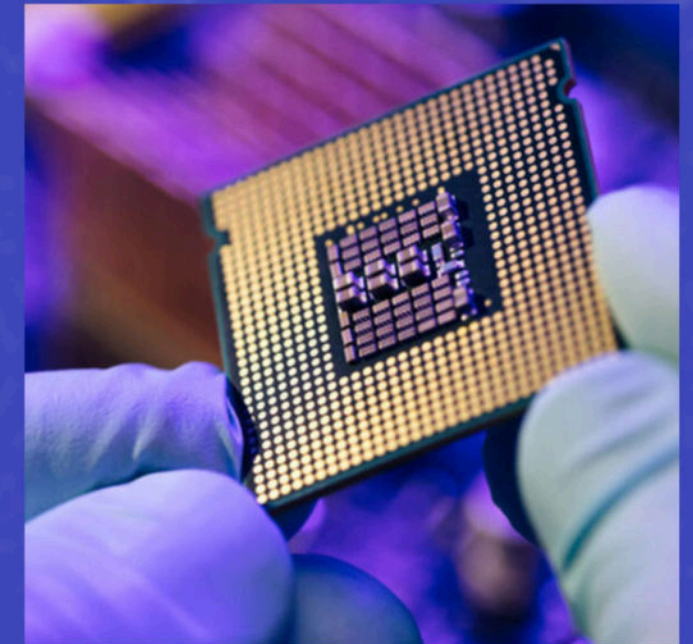


PDP-11

3ª Geração

Computadores menores,
muito mais potentes e
baratos; aumento contínuo
da velocidade e o surgimento
do Microcomputador Pessoal

Microprocessador



Apple
Lisa



4ª Geração

**Conseguem decifrar
esse código?**



**Não? isso é por não
conhecerem os símbolos utilizados
nessa representação**

**Os seres humanos se comunicam
através de códigos.**

**A língua é um código, as placas de
trânsito, os emojis e os números são
códigos.**

**Nós aprendemos a ler e
compreender esses códigos desde muito
pequenos.**

**Conseguem decifrar
esse código?**

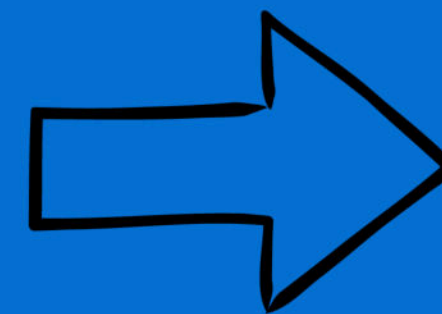


**Aqui está representado o número 151
em Kanji, um dos dialetos japoneses.**

Sistema numérico decimal

m
Kg
Km/h

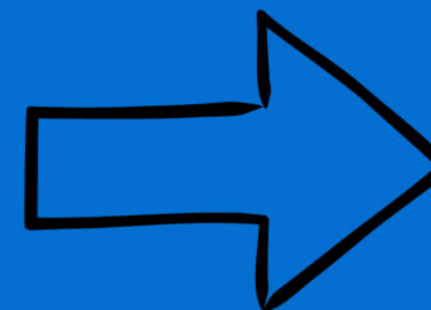
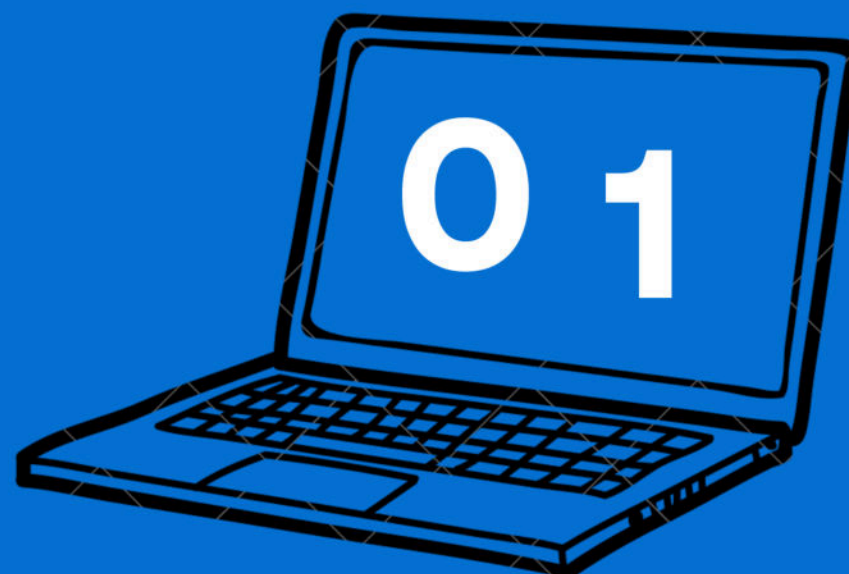
Quantidade de
algarismos



10

1 3 5 7 9
2 4 6 8 0

Infinitos
números
diferentes



2



Qual língua fala o computador?

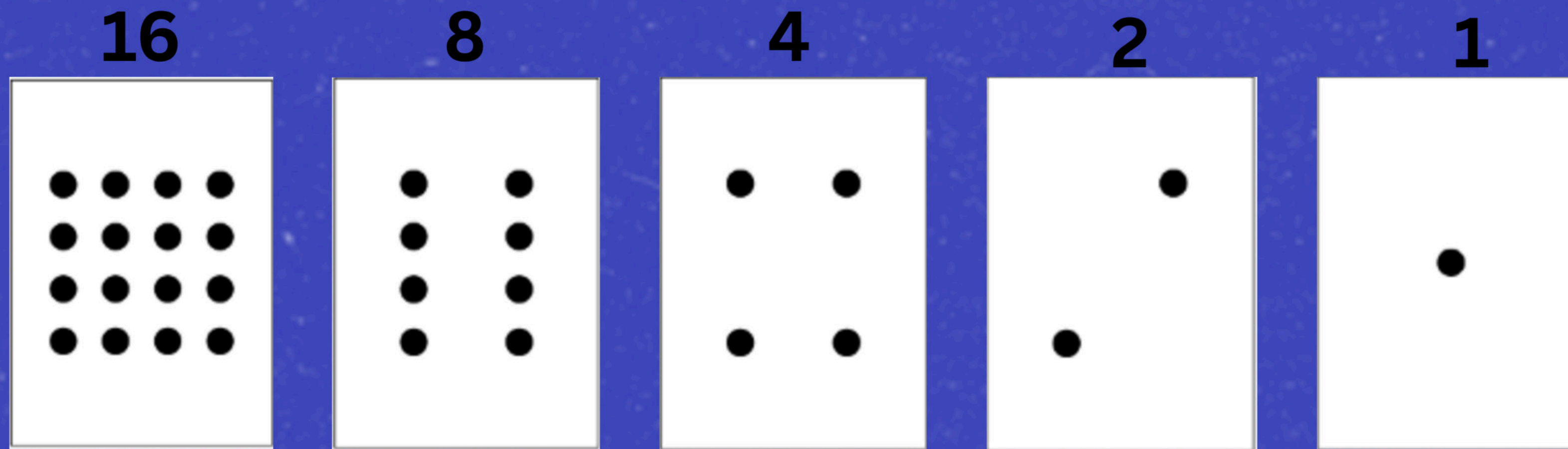
O computador compreende somente os impulsos elétricos, que podem estar ligados ou desligados. Dessa maneira, para se comunicar com as máquinas, é utilizada uma linguagem chamada Código Binário. Esse código usa somente os números 1 e 0 e consegue transmitir qualquer informação.

O computador, transforma todas as informações que inserimos nele em apenas dois números: o zero e o um.

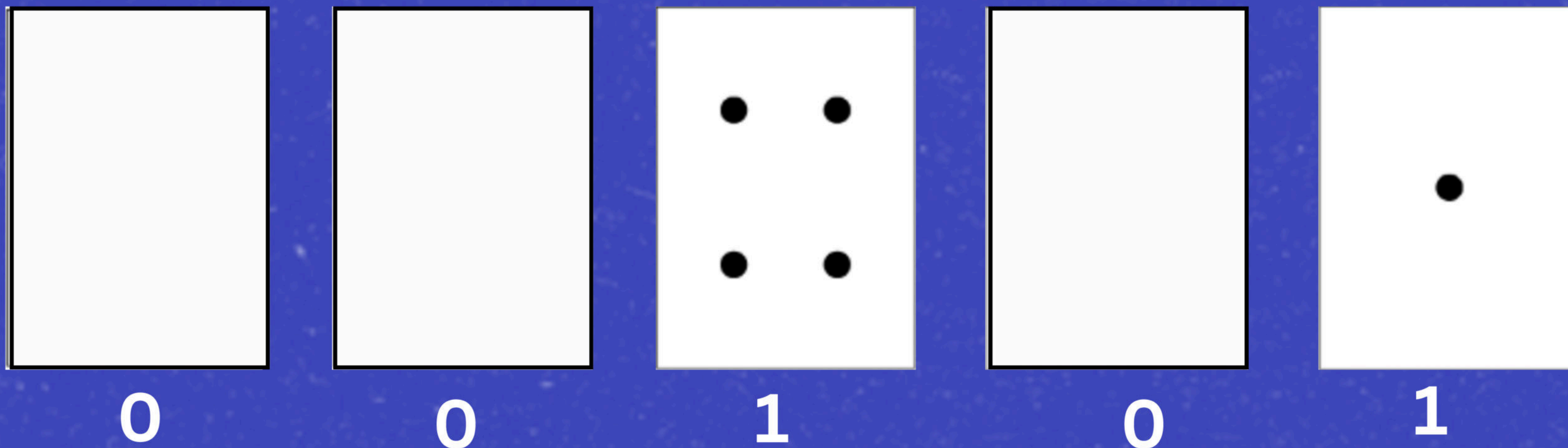
(1)  = Ligado

(0)  = Desligado

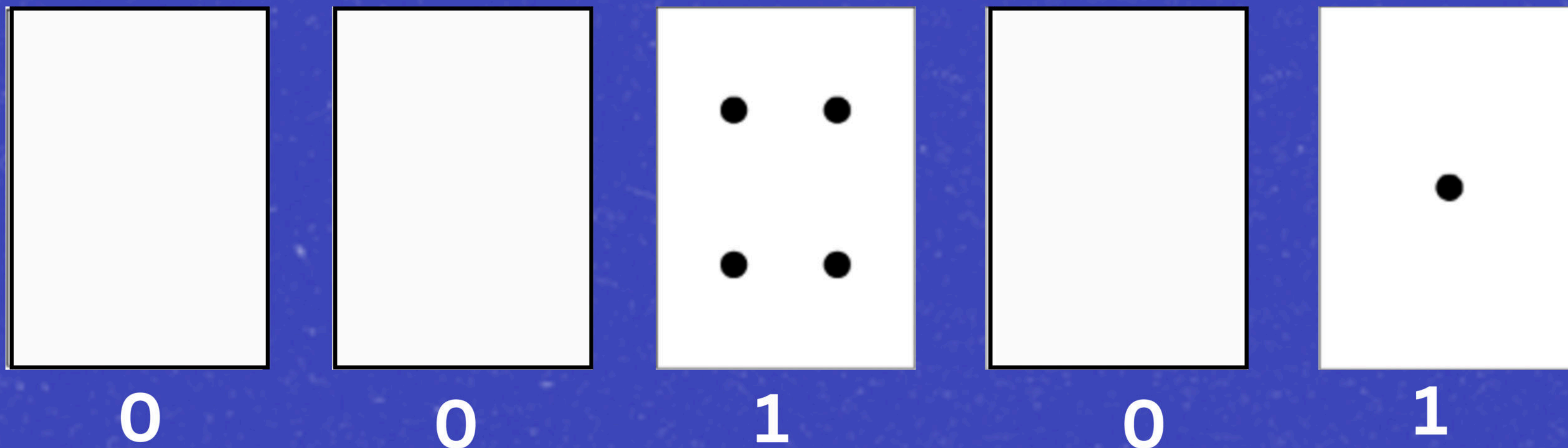
**E como converter
números para o
código binário?**



Caso a carta esteja em branco, o computador entende como 0, caso esteja com os pontos exibidos, então ele entende como 1



O computador faz a soma dos pontos das cartas com a face pra cima (que ele entende como 1), qual será o número que o computador vai entender?



Se você falou 5, acertou!

Diagram illustrating the addition of 4 and 1 to get 5. On the left, a box contains four dots arranged in a 2x2 grid. This is followed by a plus sign (+). To the right of the plus sign is a box containing one dot in the center. This is followed by an equals sign (=). To the right of the equals sign is the number 5.

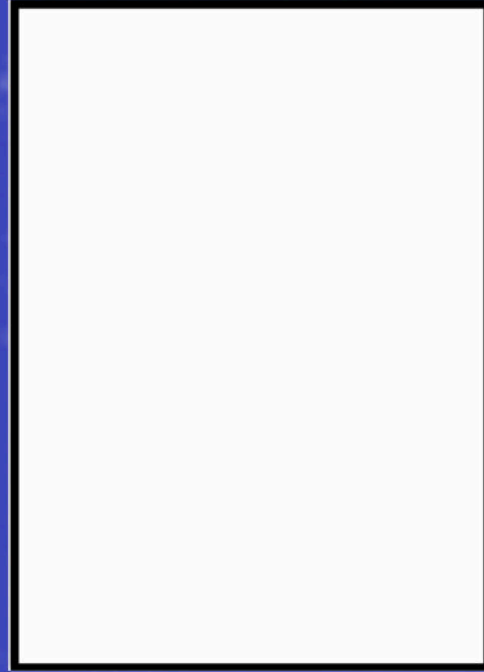
**O número 5 é
representado em código
binário por: 0101**

Vamos practicar??

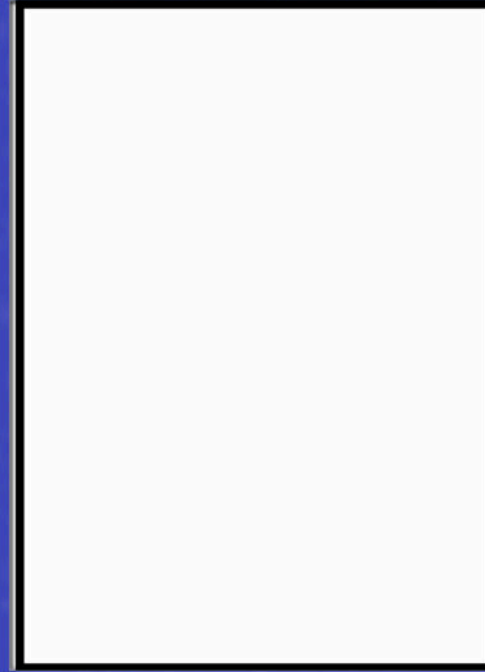
Qual número o código abaixo representa?



0



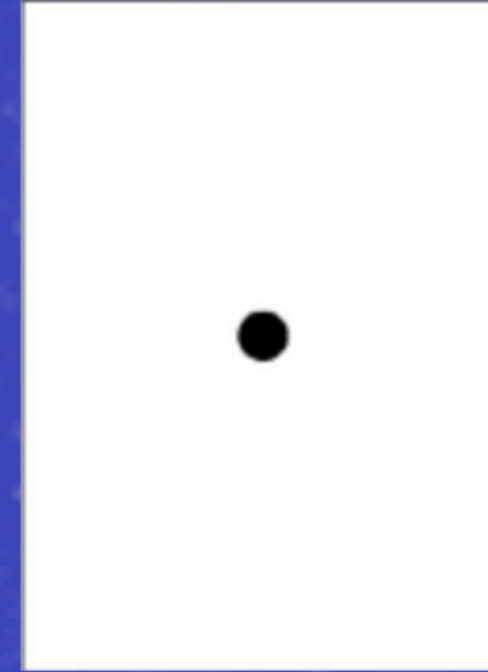
0



0

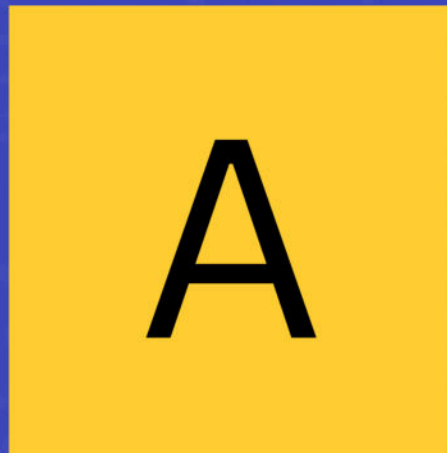


1



1

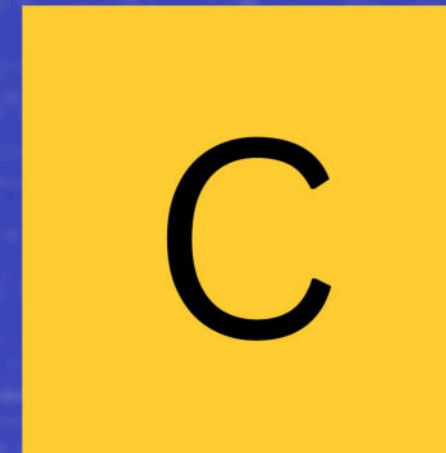
2



8



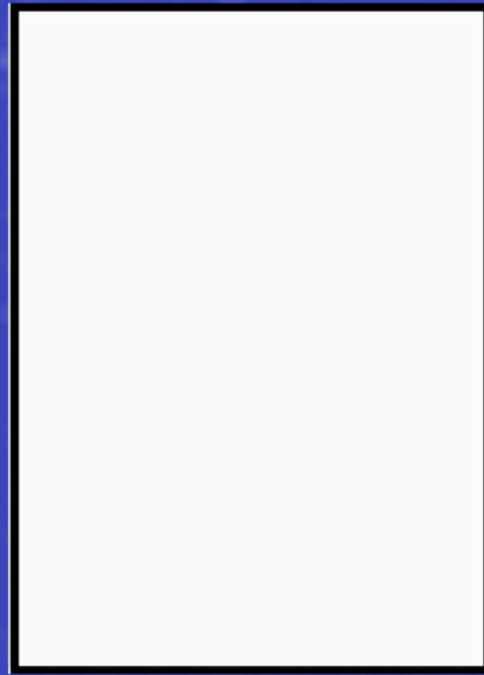
3



Qual número o código abaixo representa?



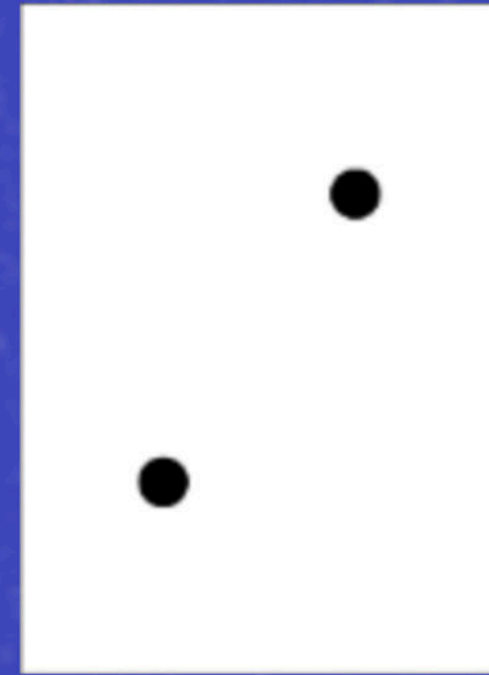
0



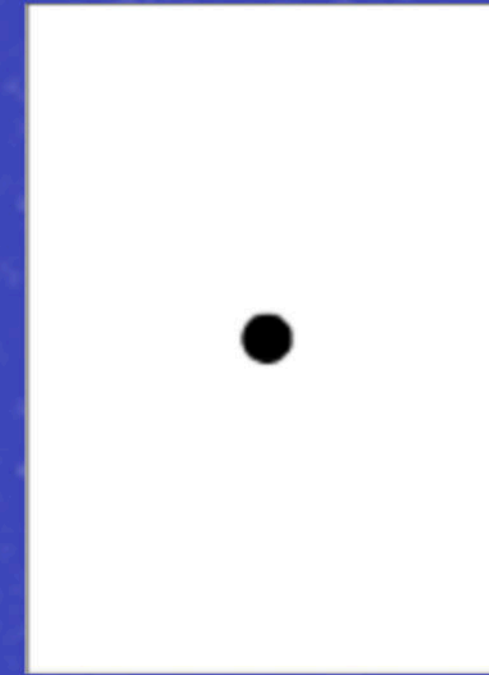
0



0



1

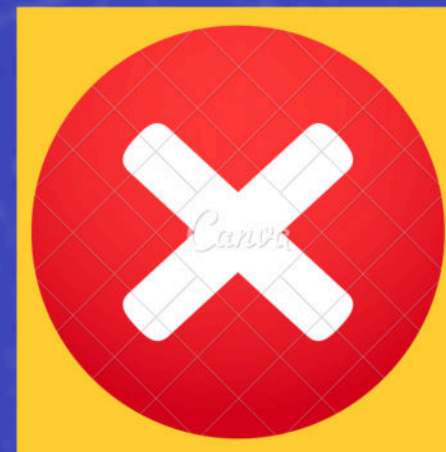


1

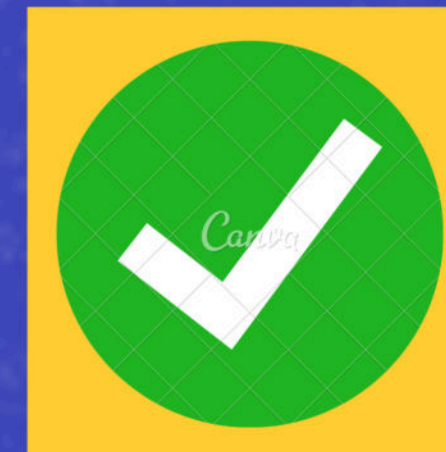
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8



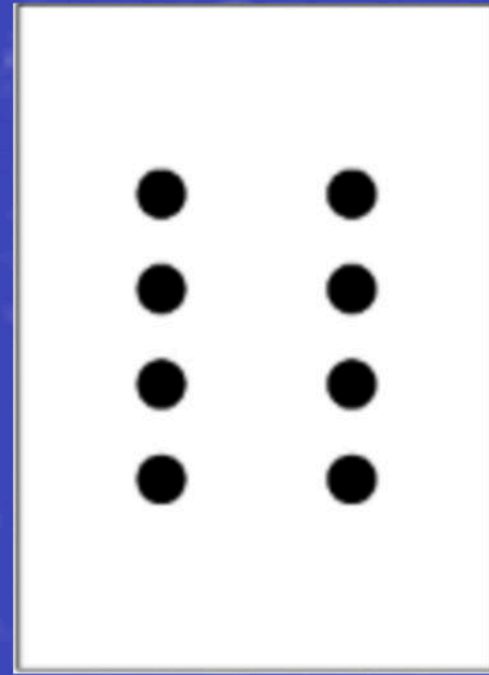
3



Qual número o código abaixo representa?



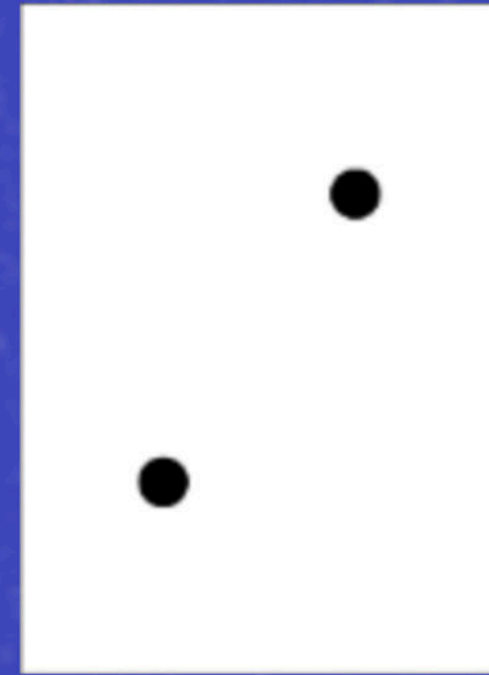
0



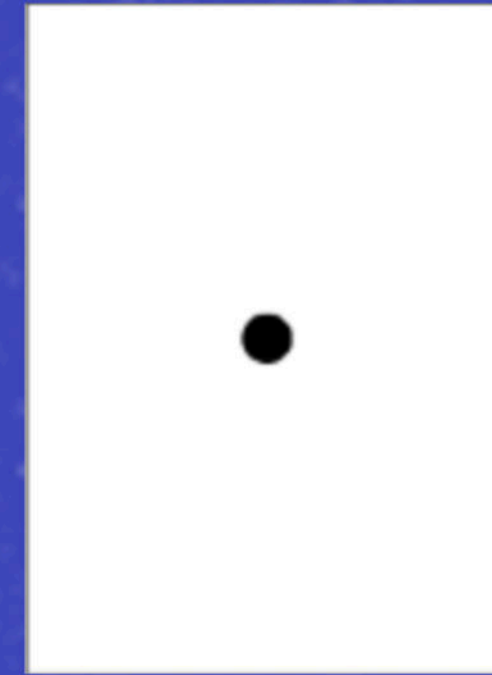
1



0



1



1

11

A

5

B

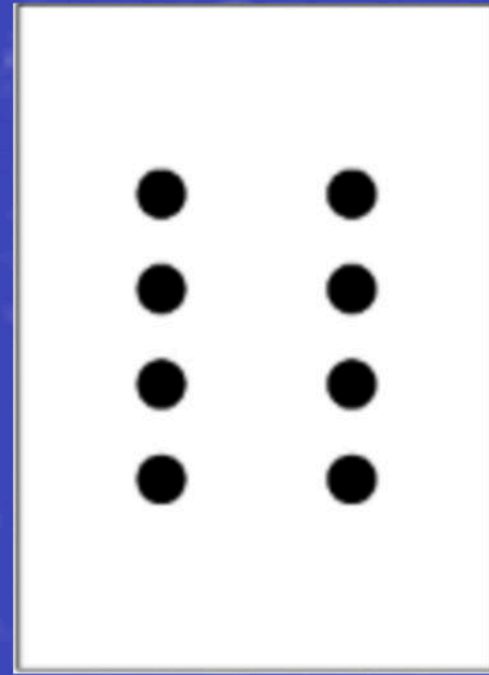
8

C

Qual número o código abaixo representa?



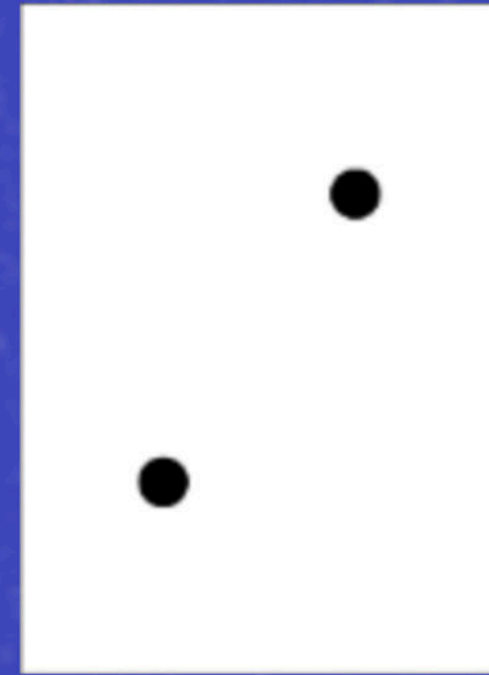
0



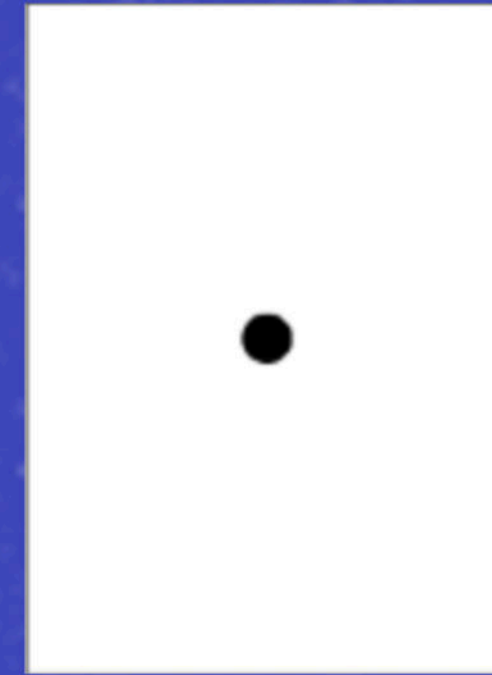
1



0

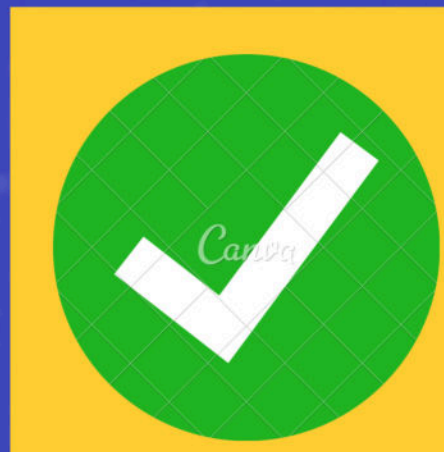


1

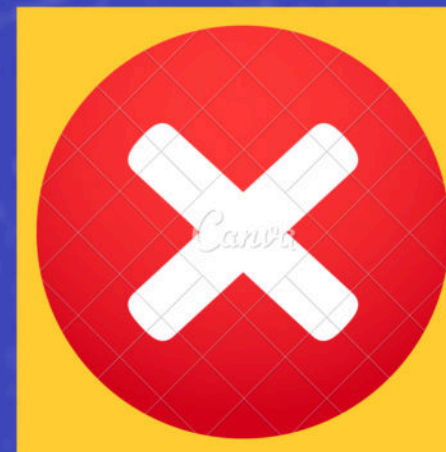


1

11



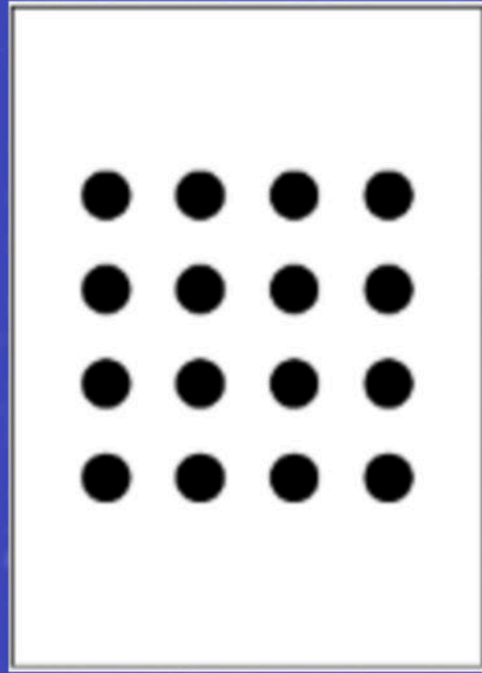
5



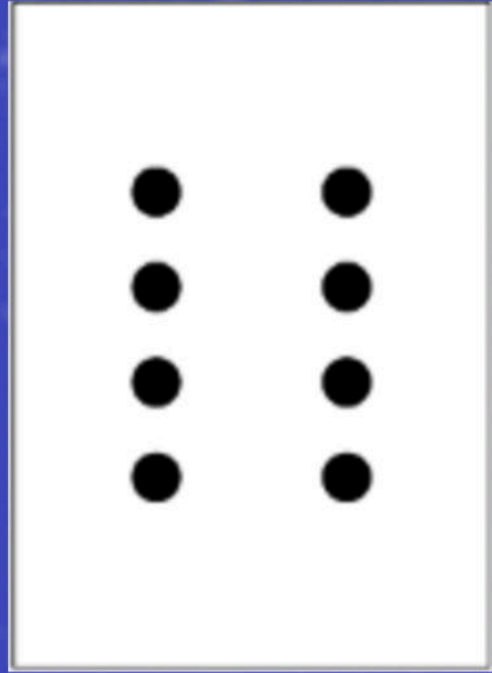
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Qual número o código abaixo representa?



1



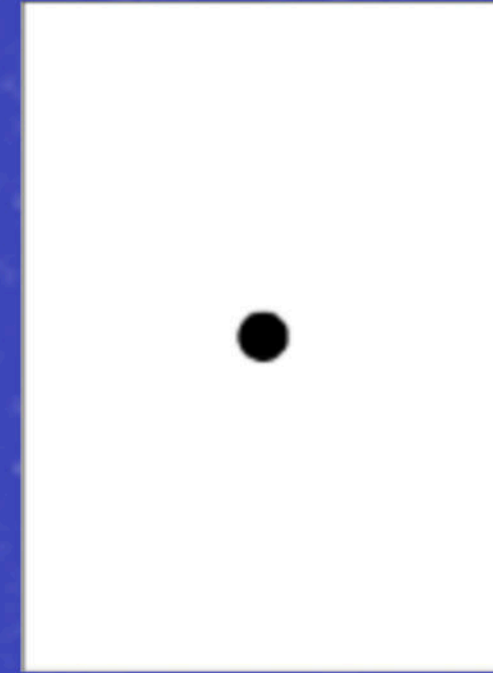
1



0



0



1

17

A

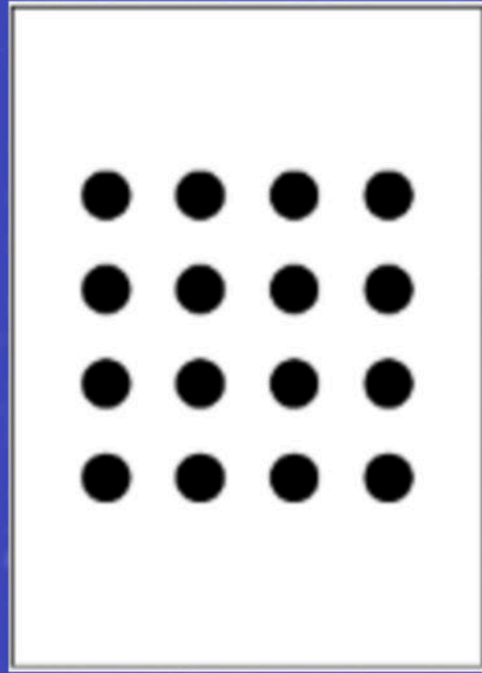
15

B

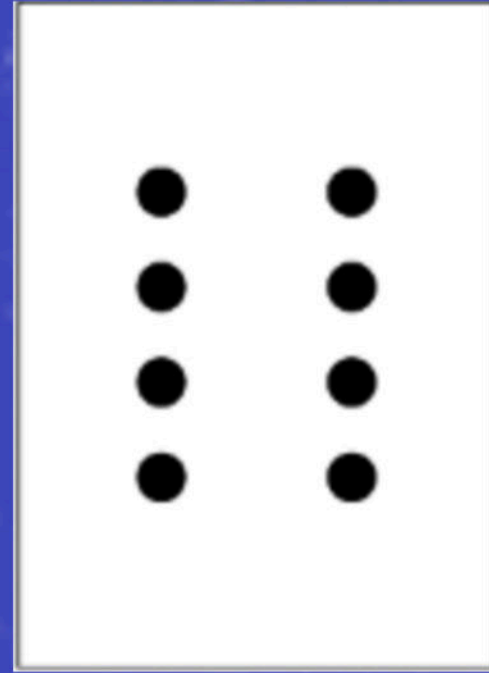
25

C

Qual número o código abaixo representa?



1



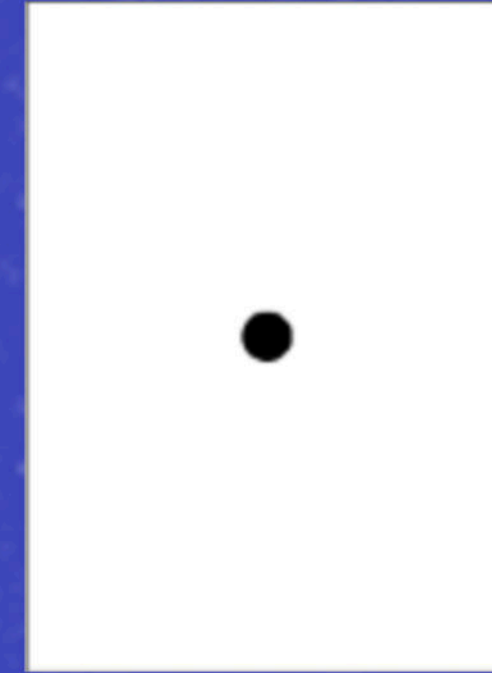
1



0

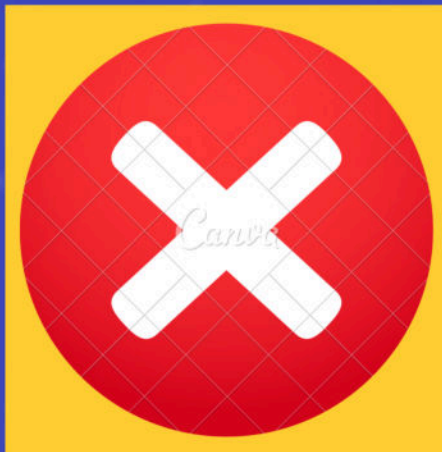


0



1

17



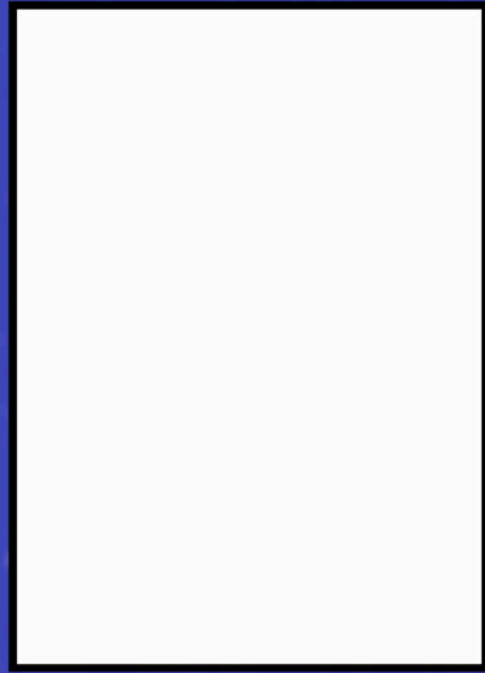
15



25



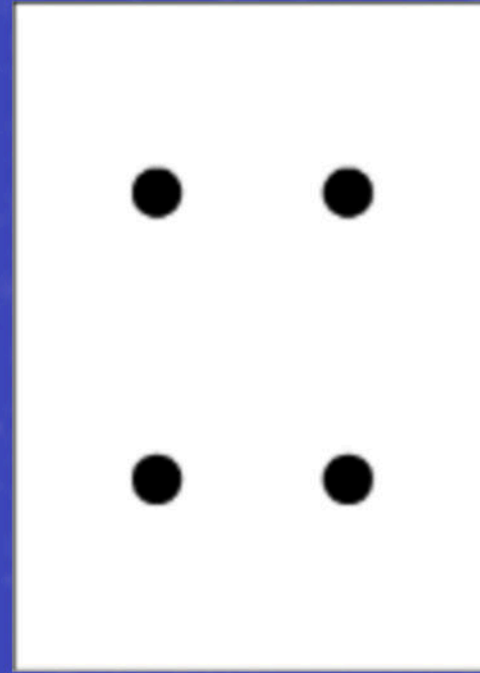
Qual o código que representa o número 6?



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0110

A

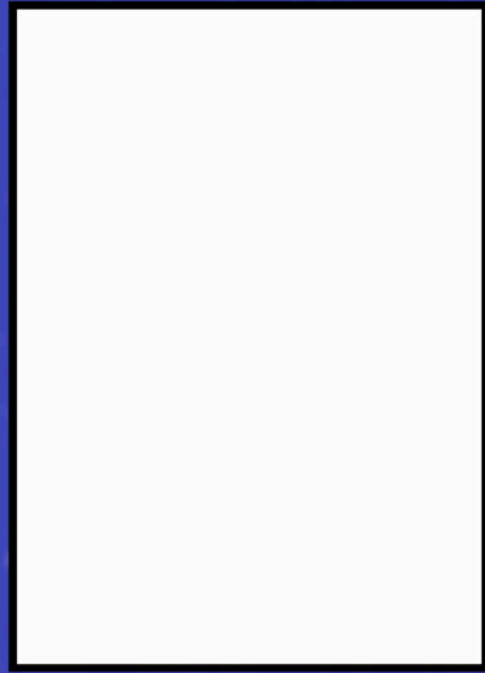
0010

B

0111

C

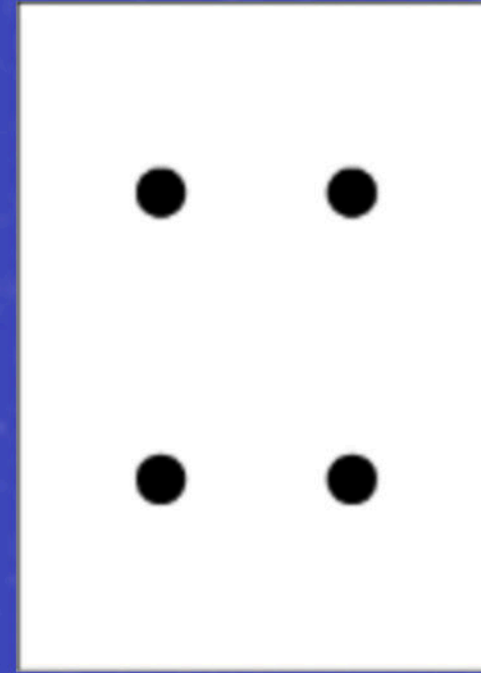
Qual o código que representa o número 6?



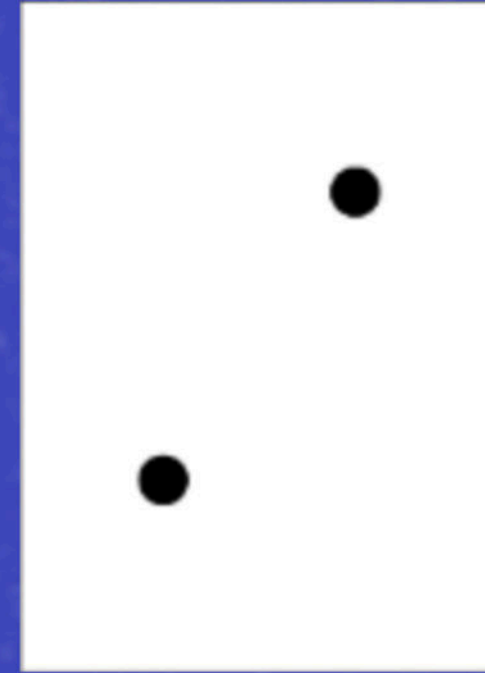
?



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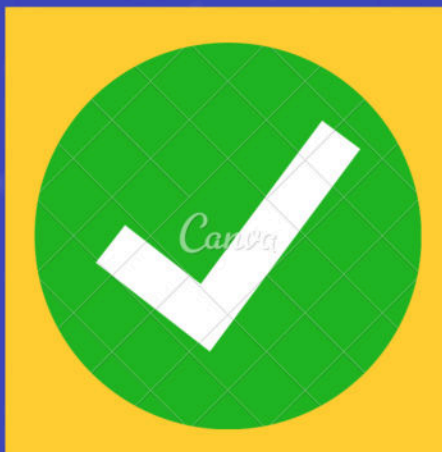


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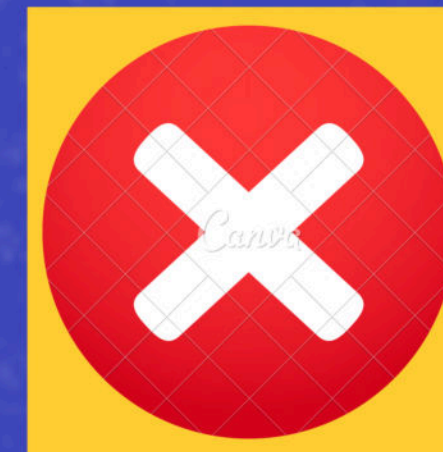
0110



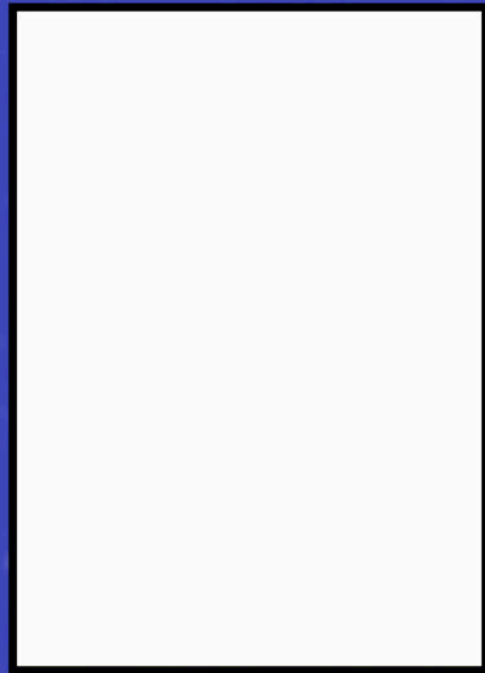
0010



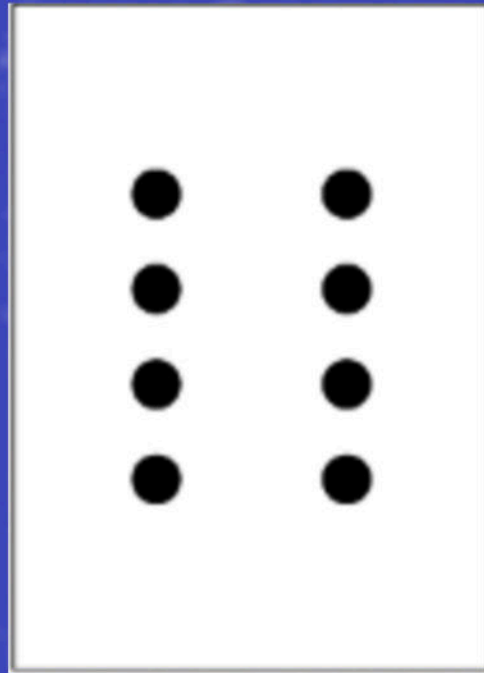
0111



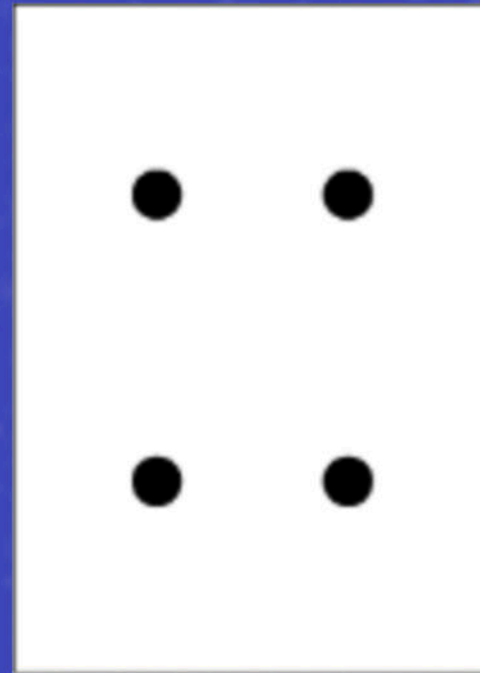
Qual o código que representa o número 14?



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11010

A

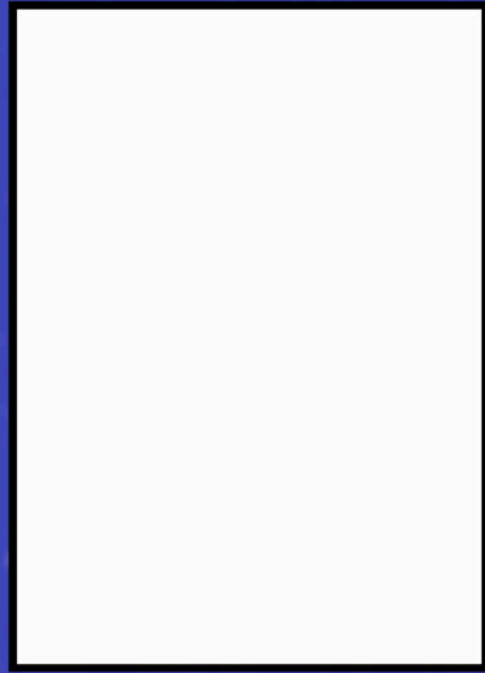
01110

B

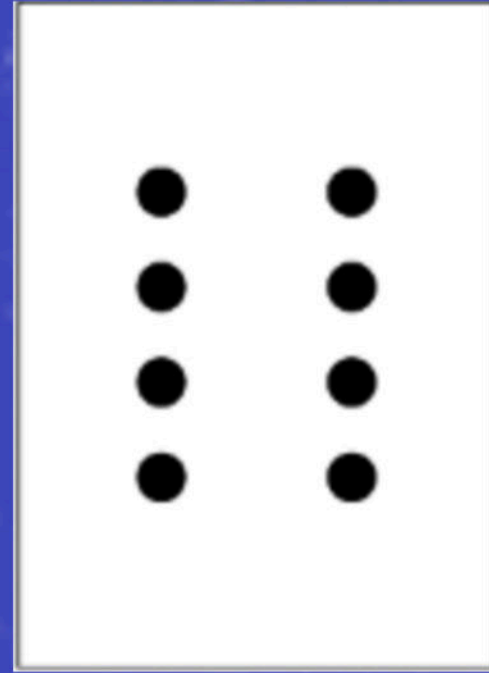
00110

C

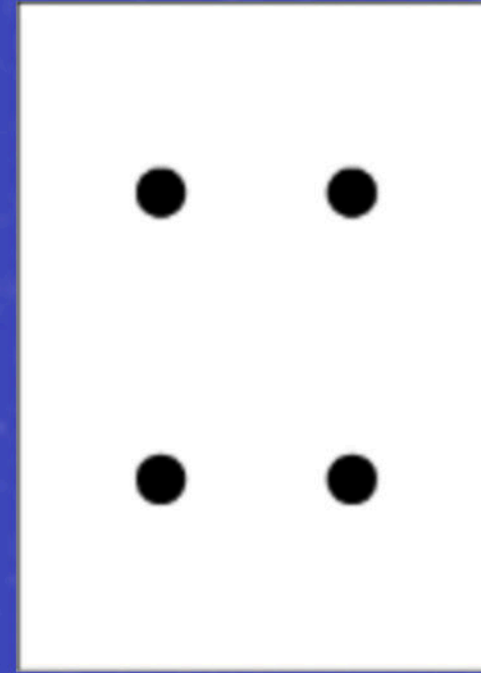
Qual o código que representa o número 14?



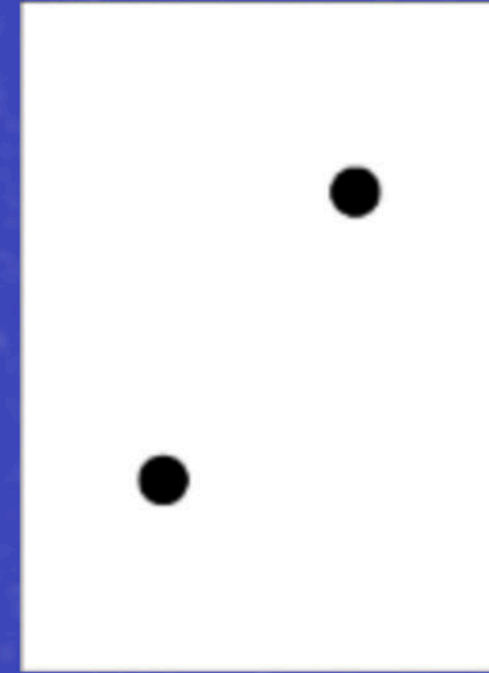
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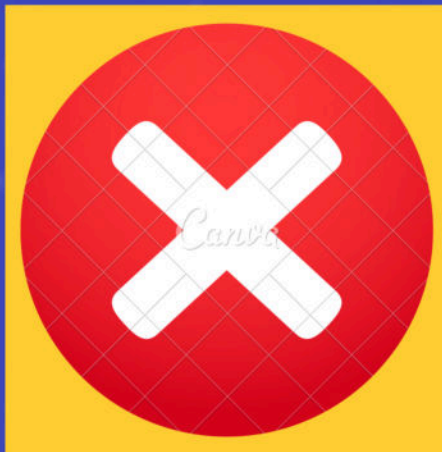


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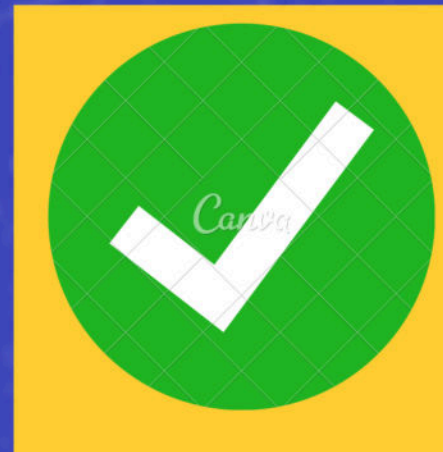


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11010



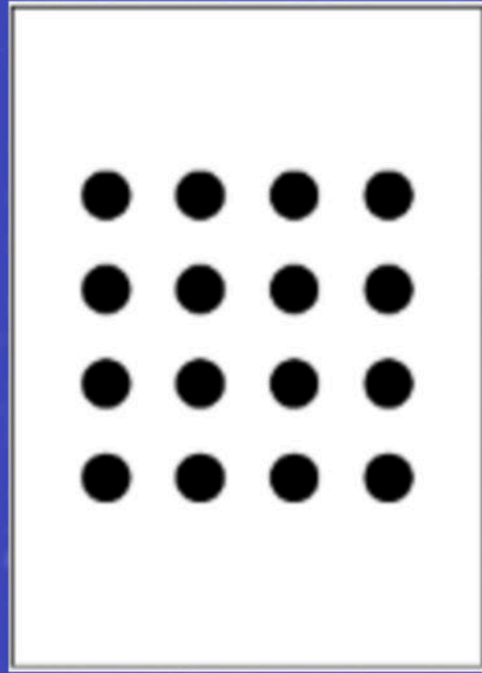
01110



00110



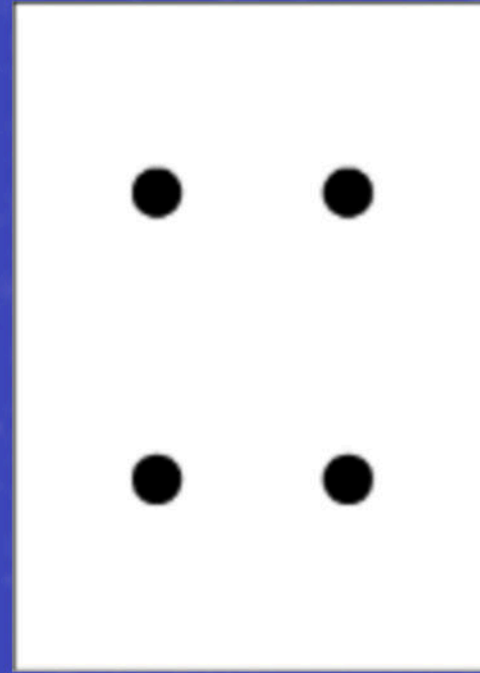
Qual o código que representa o número 22?



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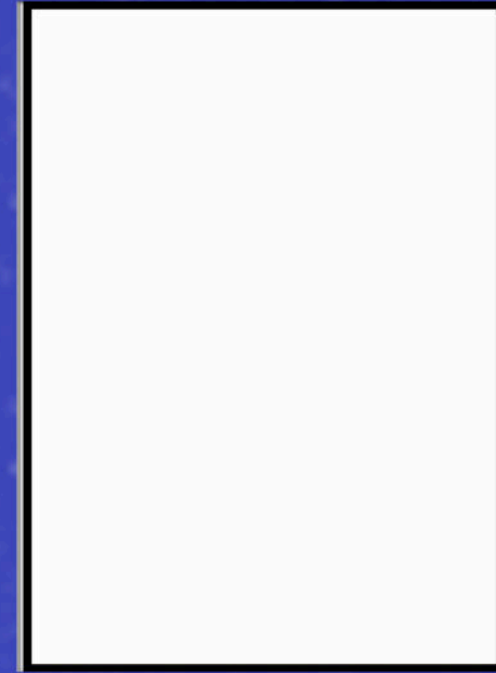
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10110

A

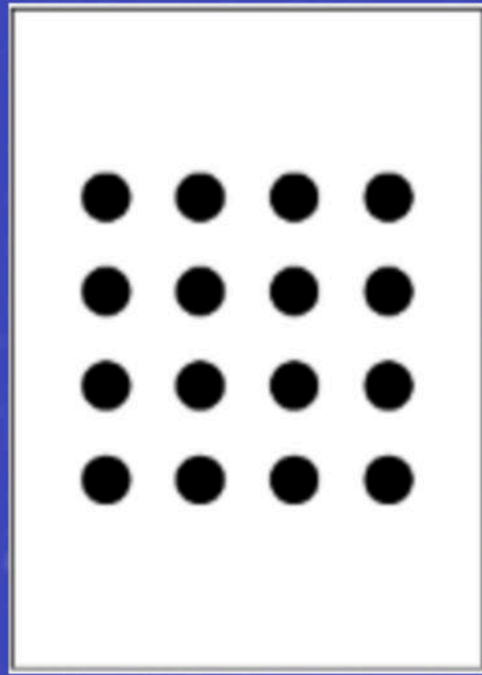
00110

B

10100

C

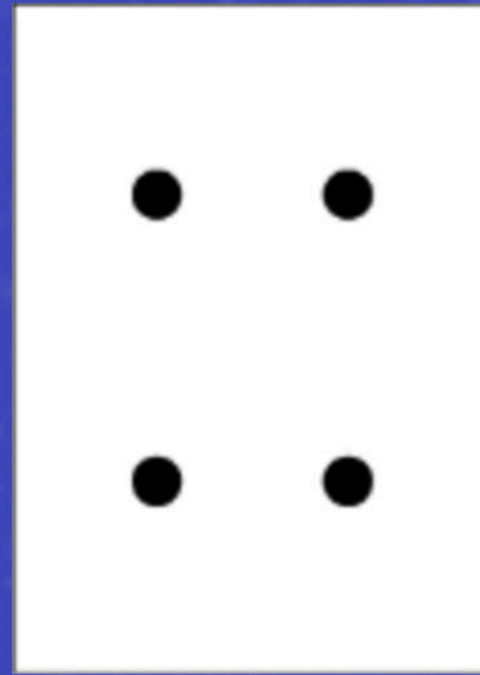
Qual o código que representa o número 22?



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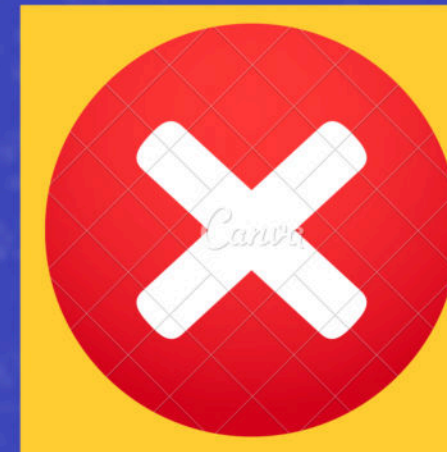
10110



00110

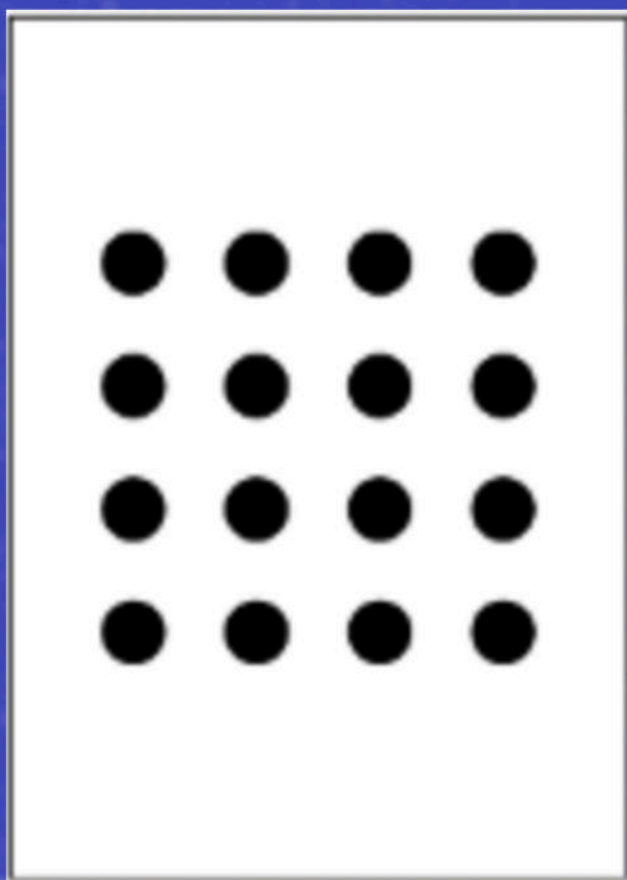


10100

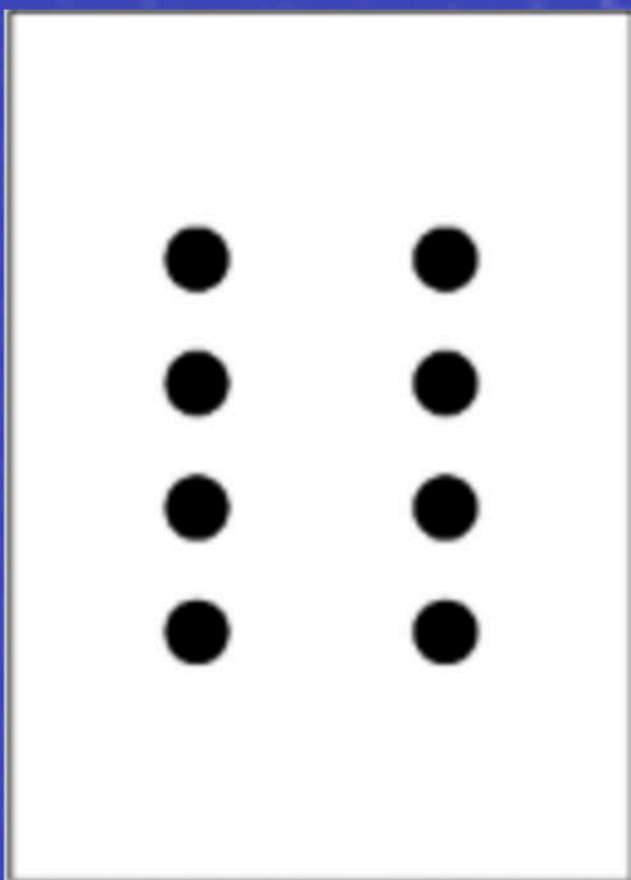


Preciso de 5 alunos !

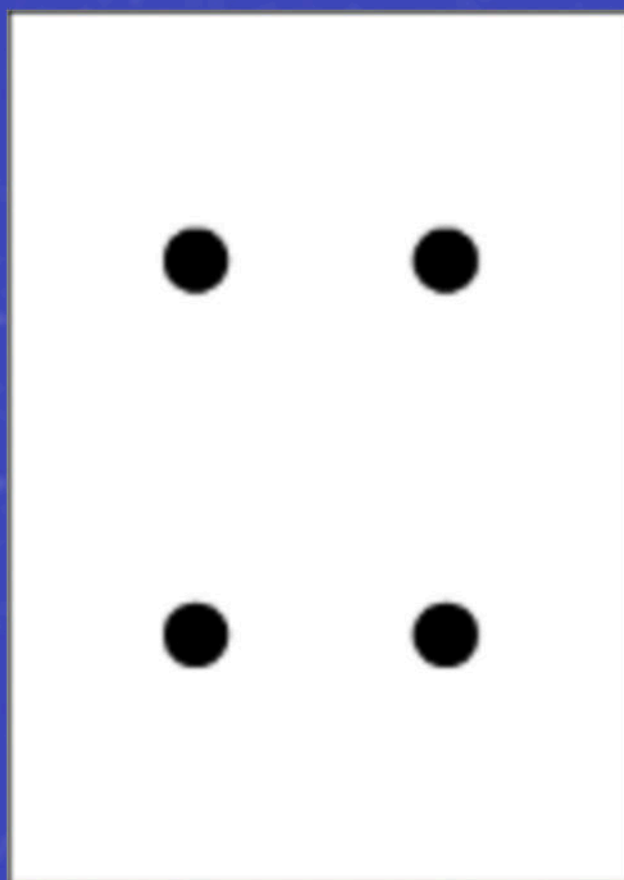
16



8



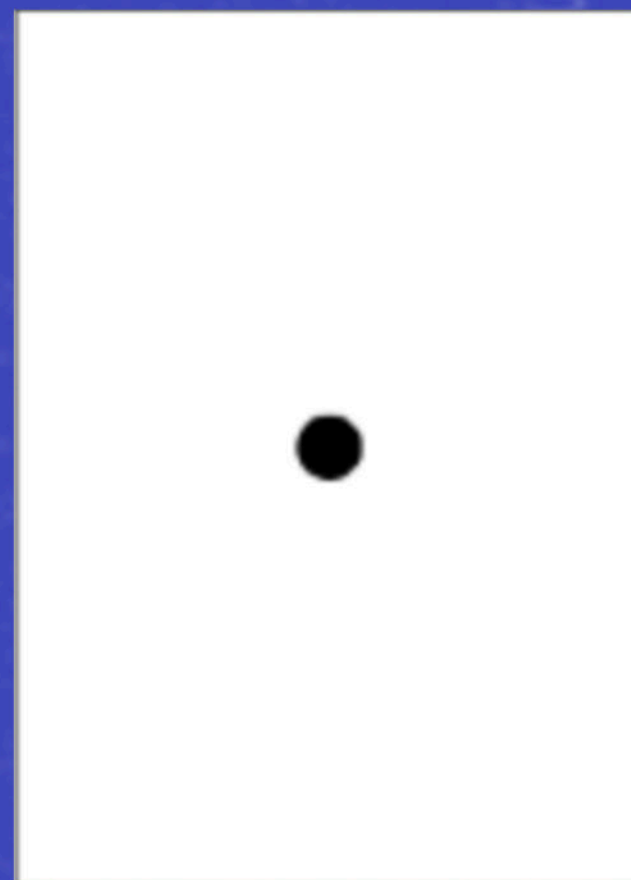
4



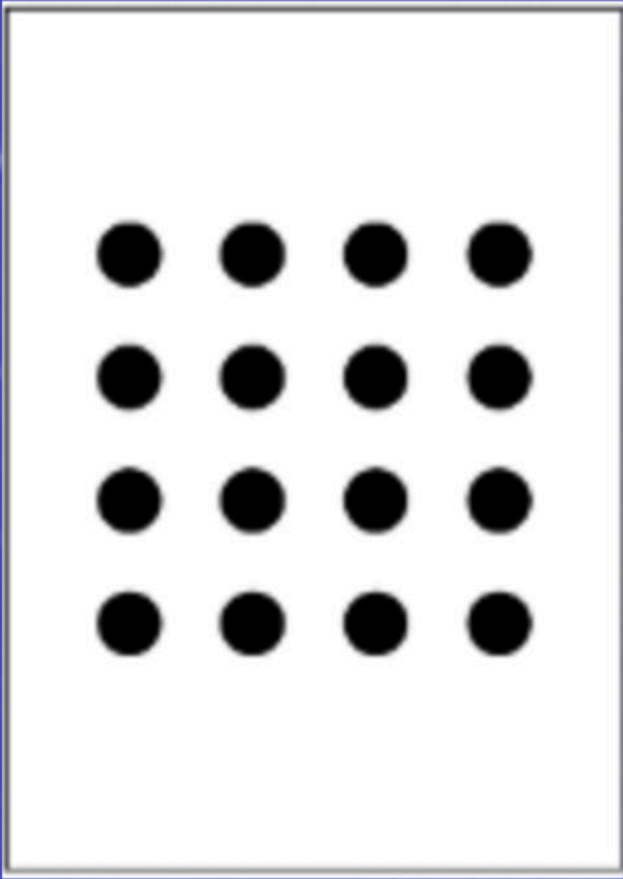
2



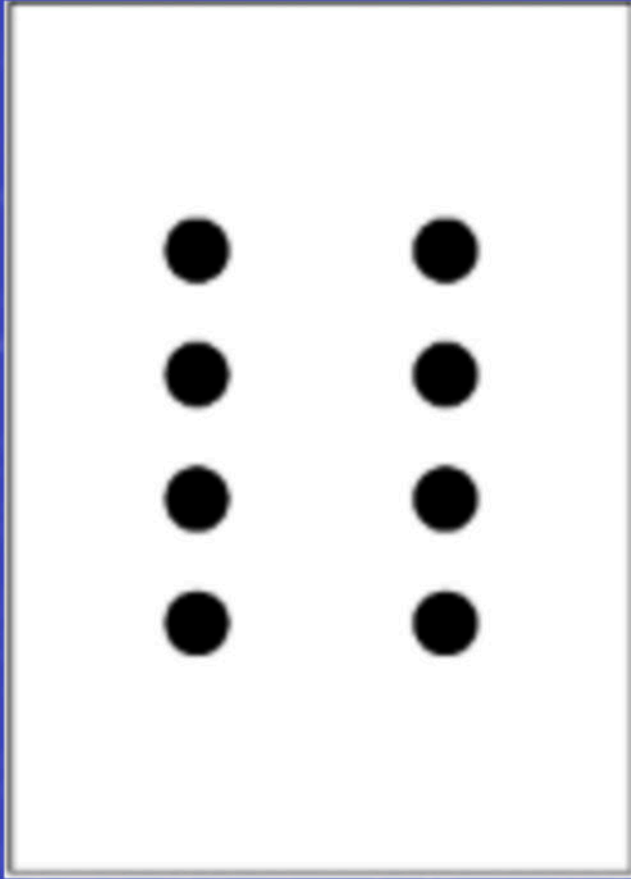
1



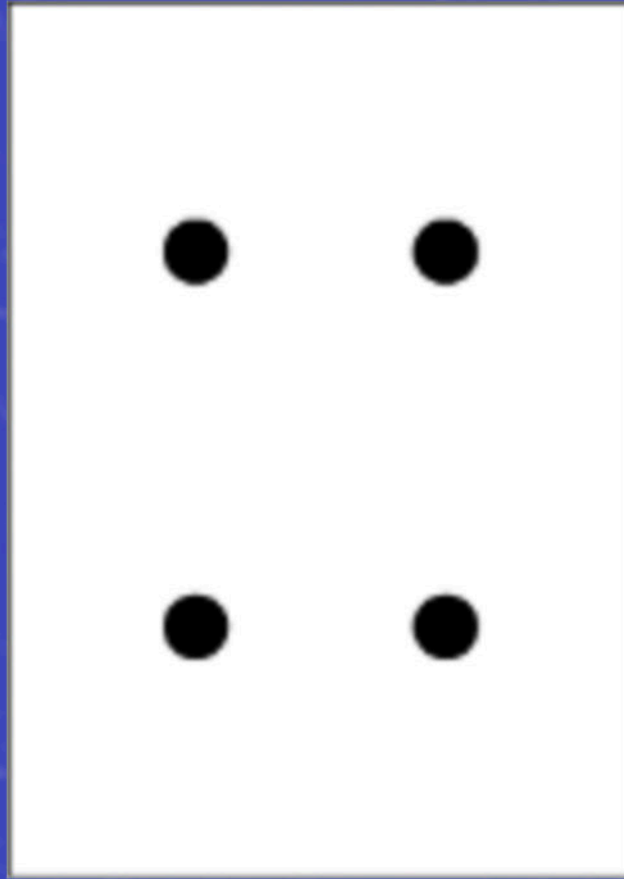
16



8



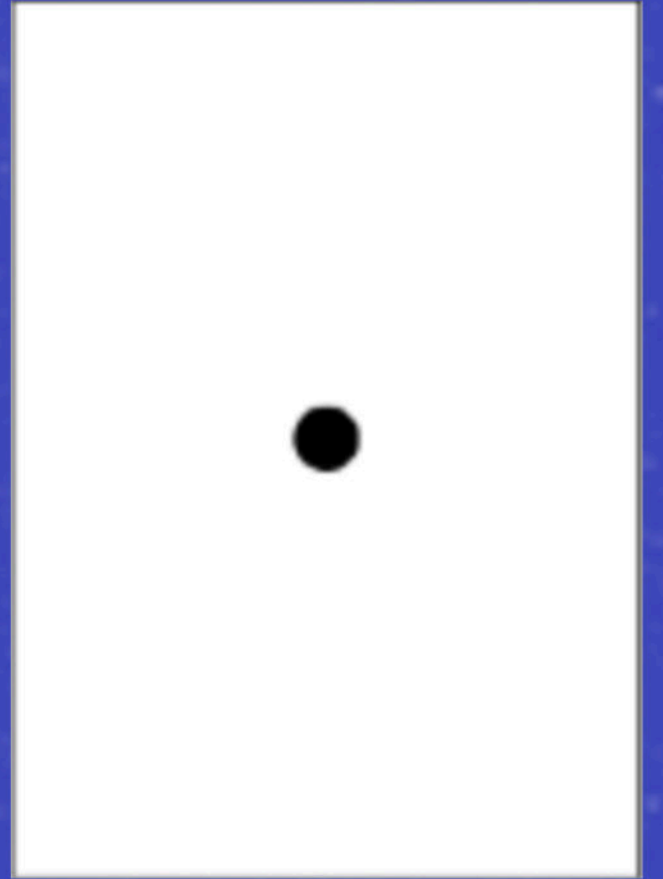
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2

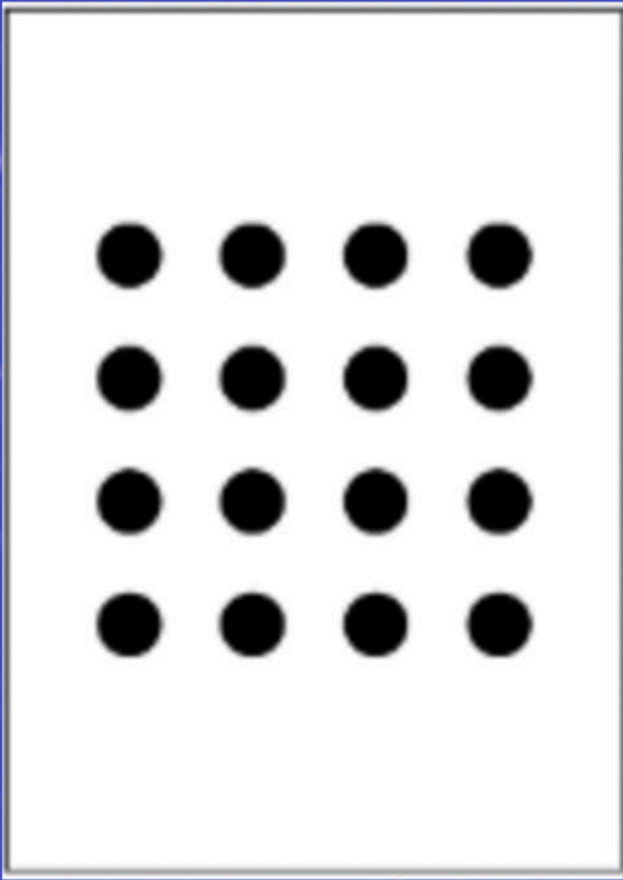


1

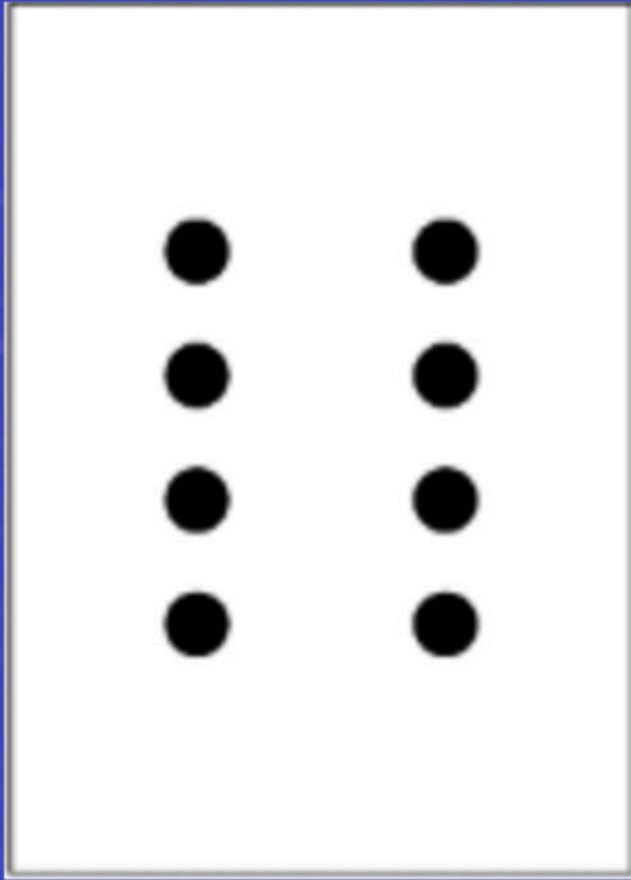


5 =

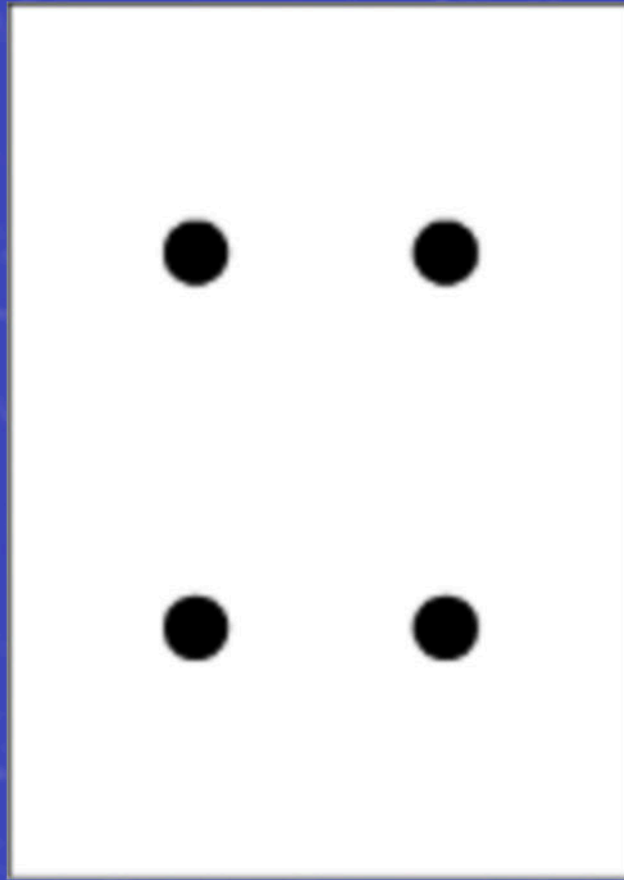
16



8



4



2

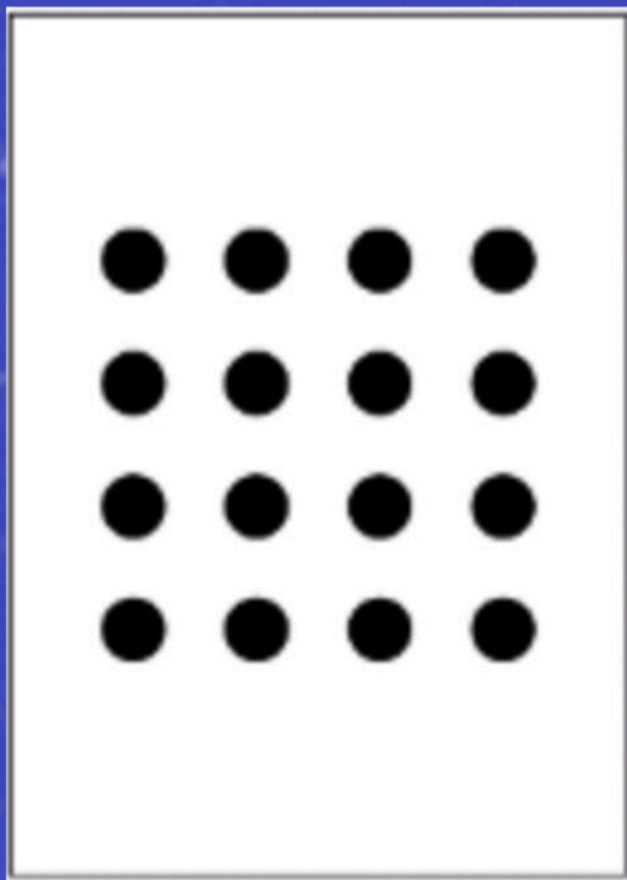


1

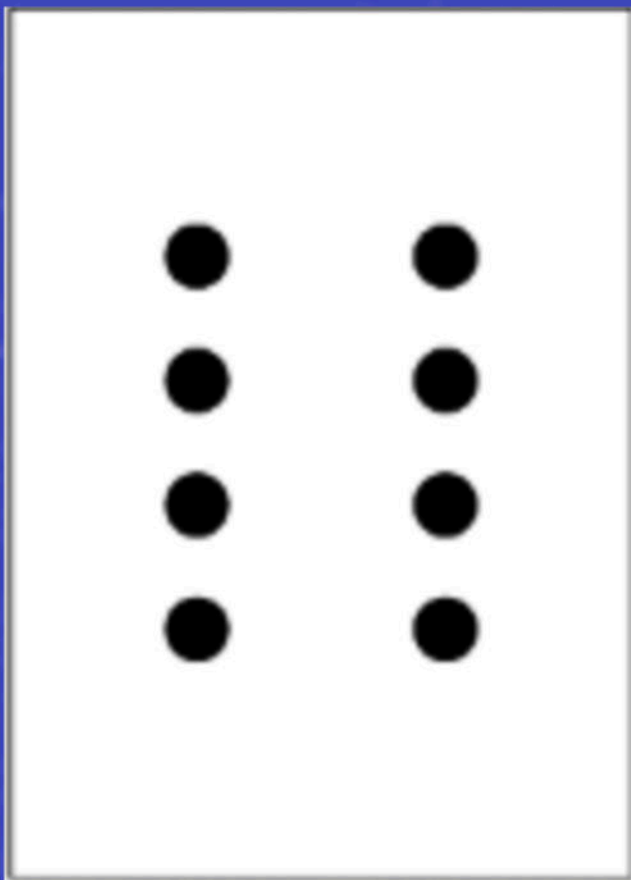


5 = 0101

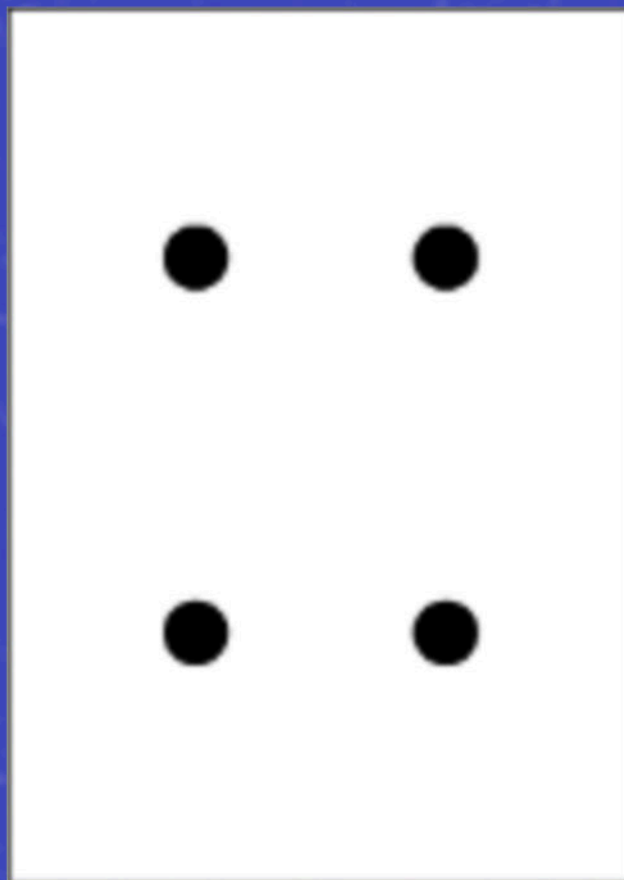
16



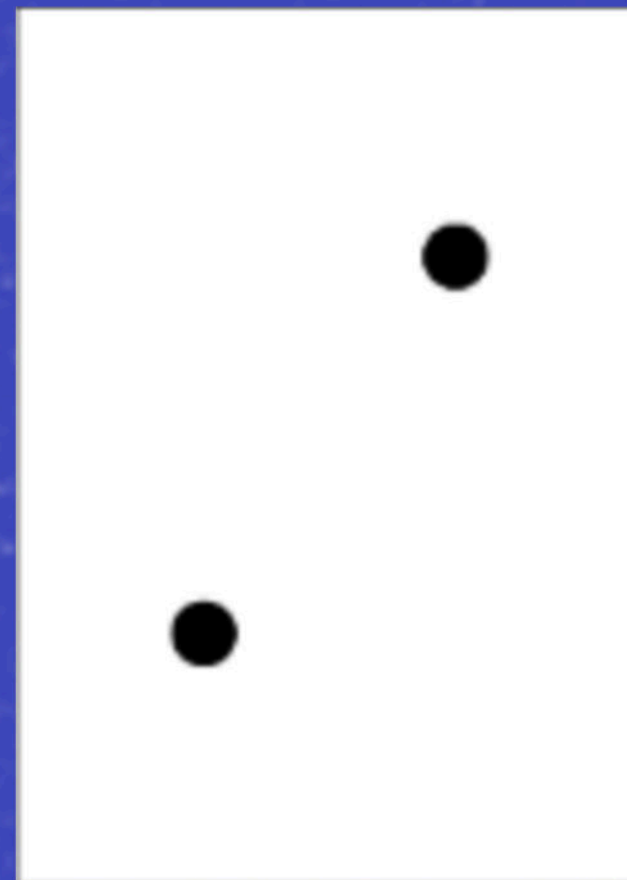
8



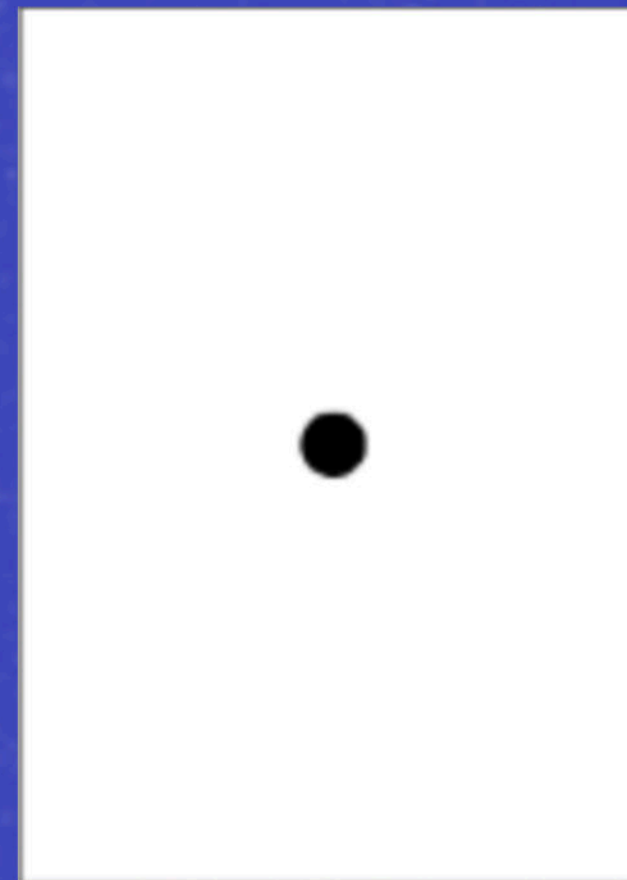
4



2



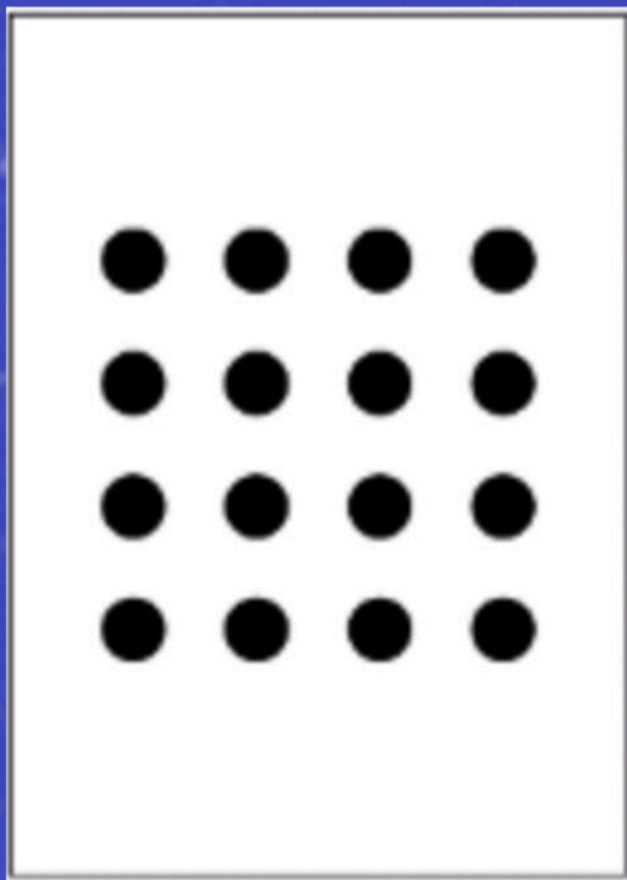
1



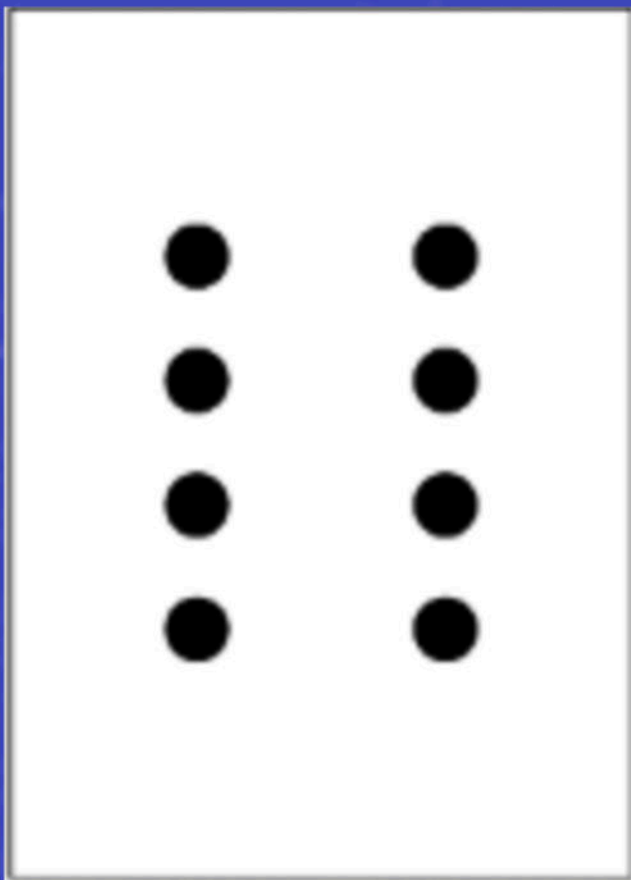
5 = 0101

20 =

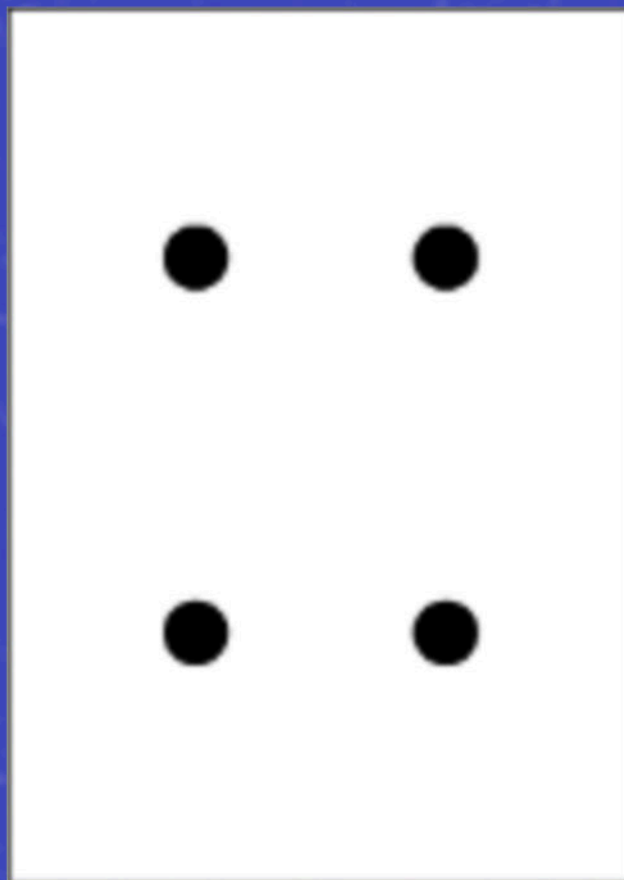
16



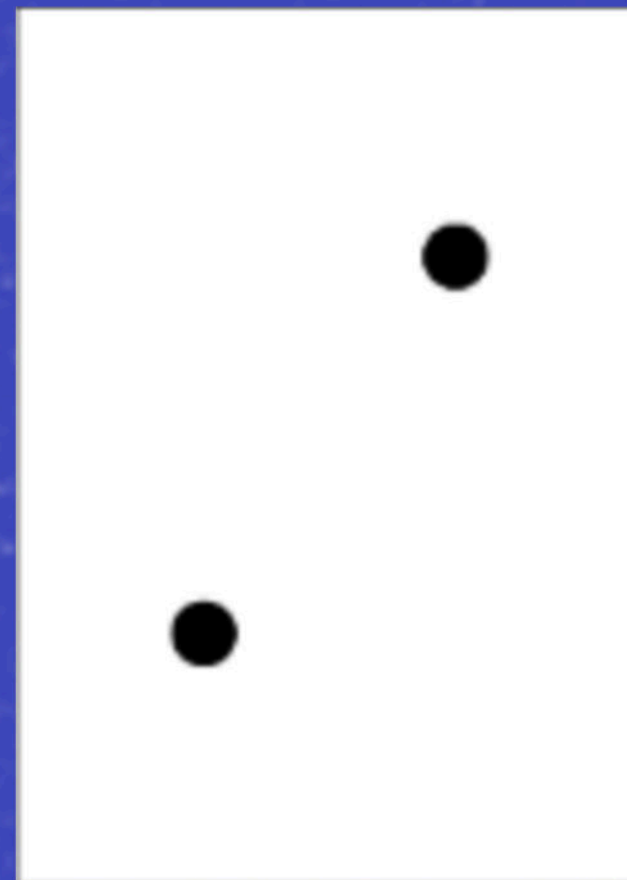
8



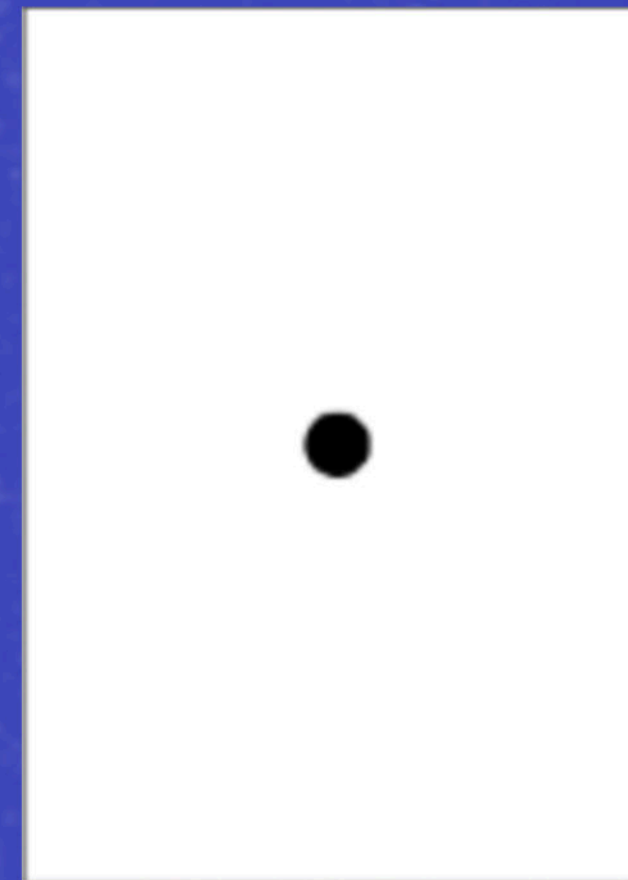
4



2



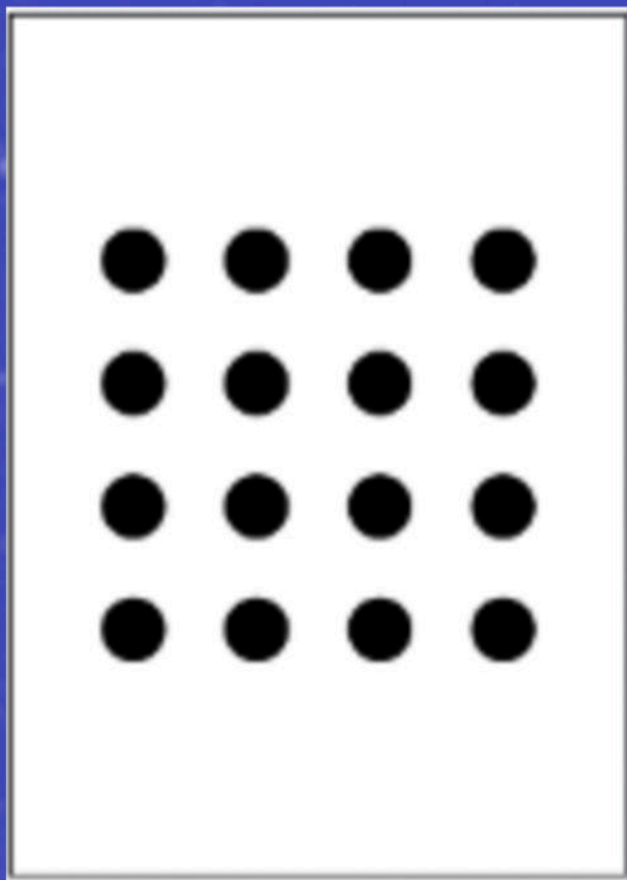
1



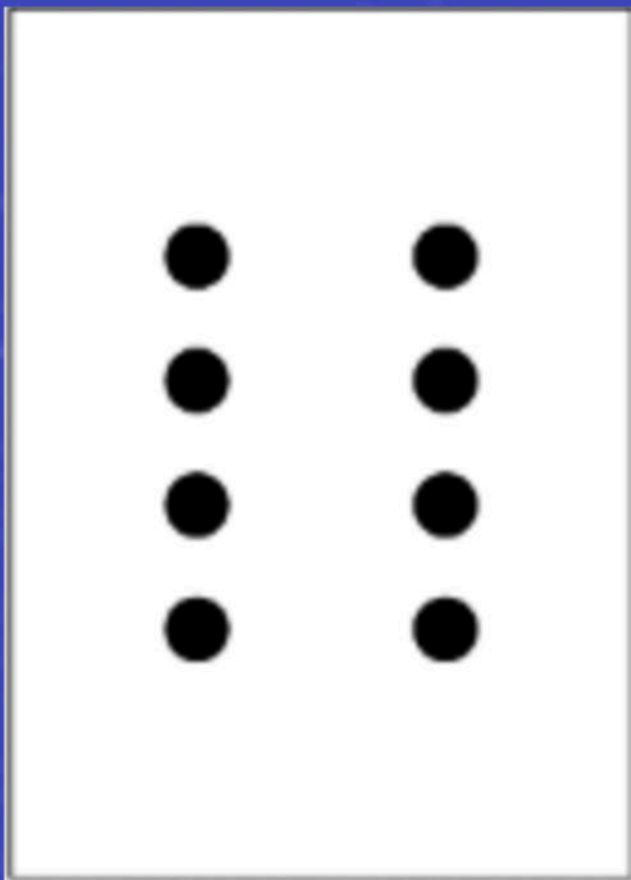
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20 = 1010

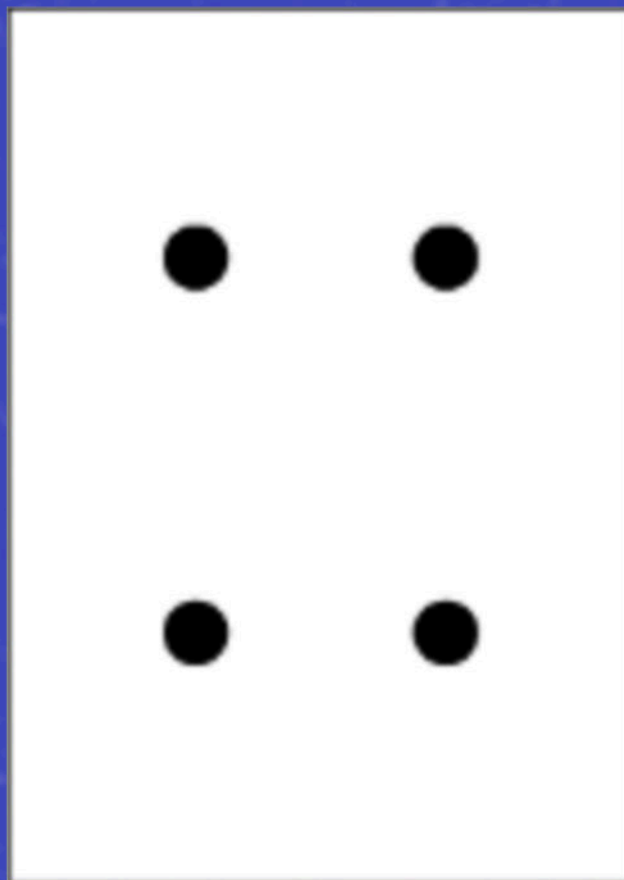
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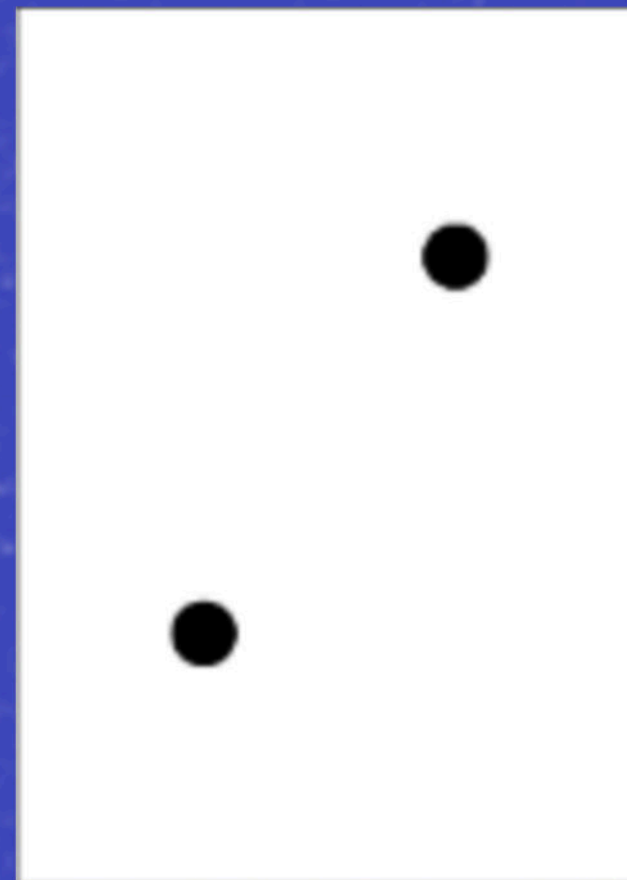
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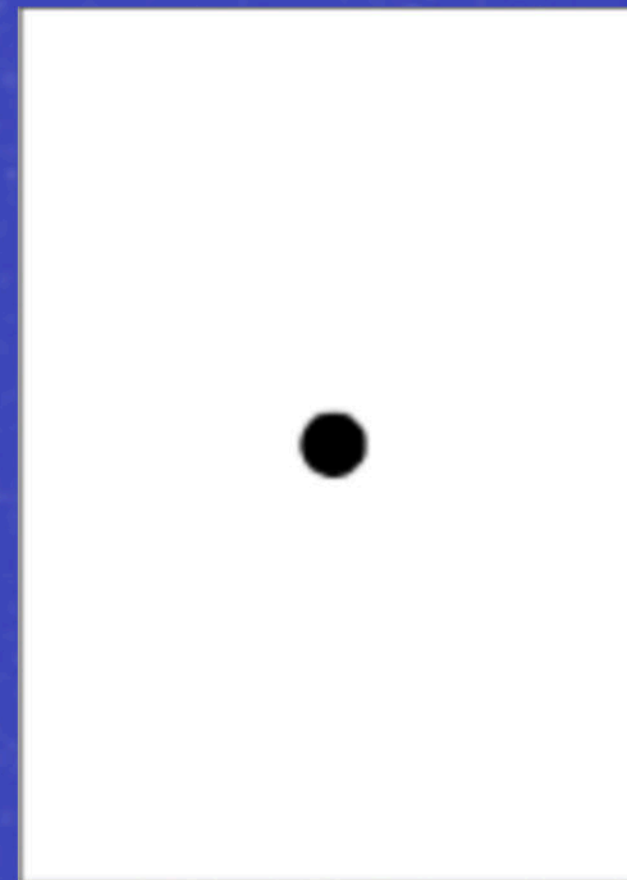
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2



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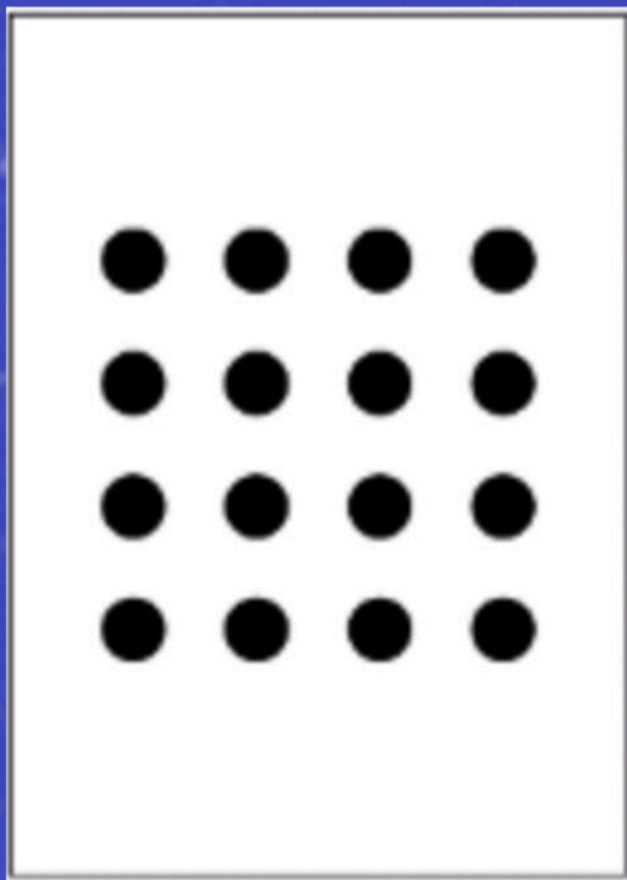


5 = 0101

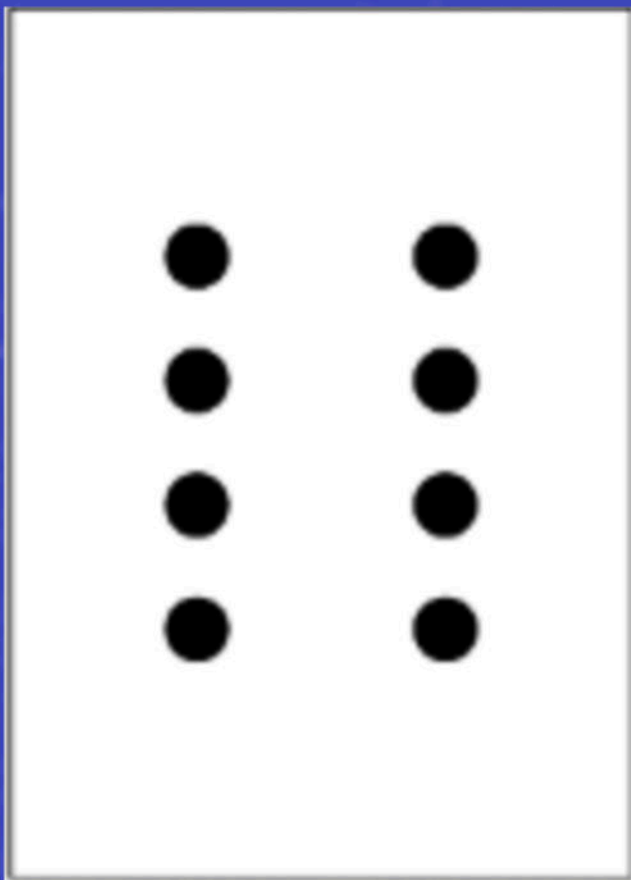
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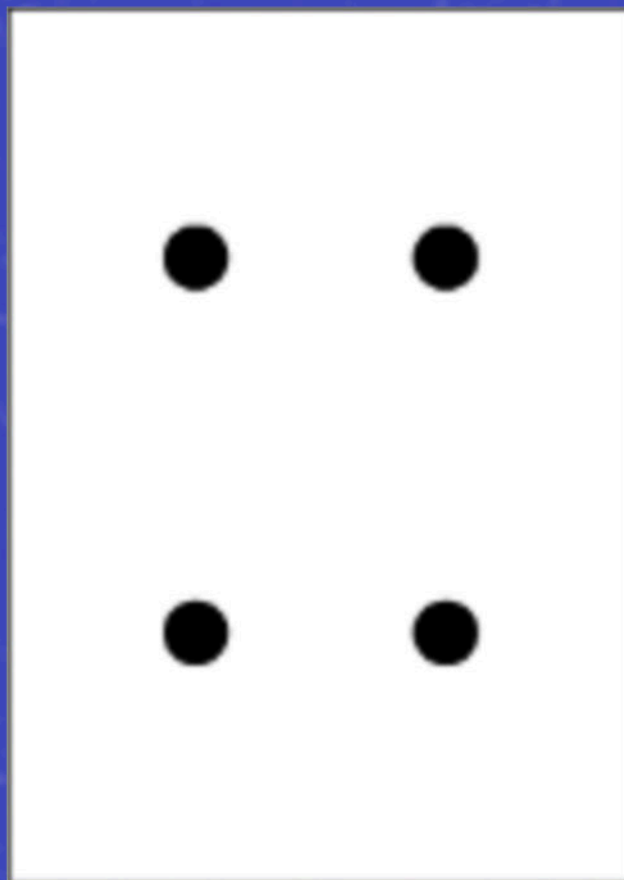
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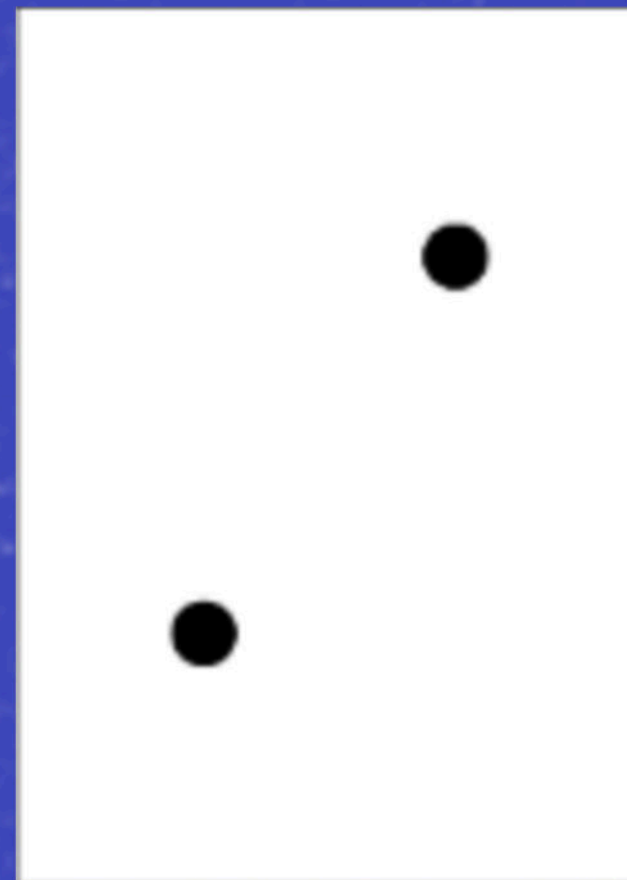
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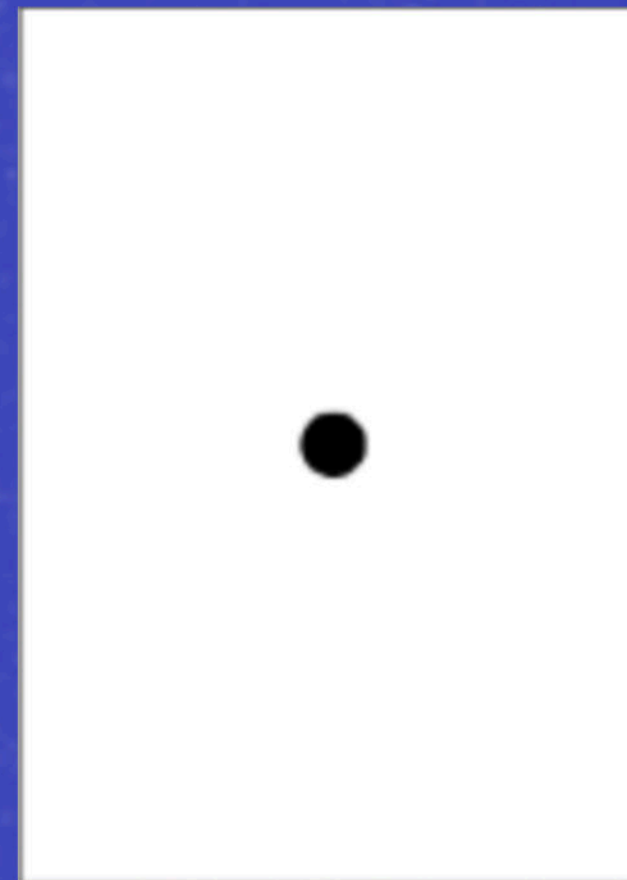
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2



1

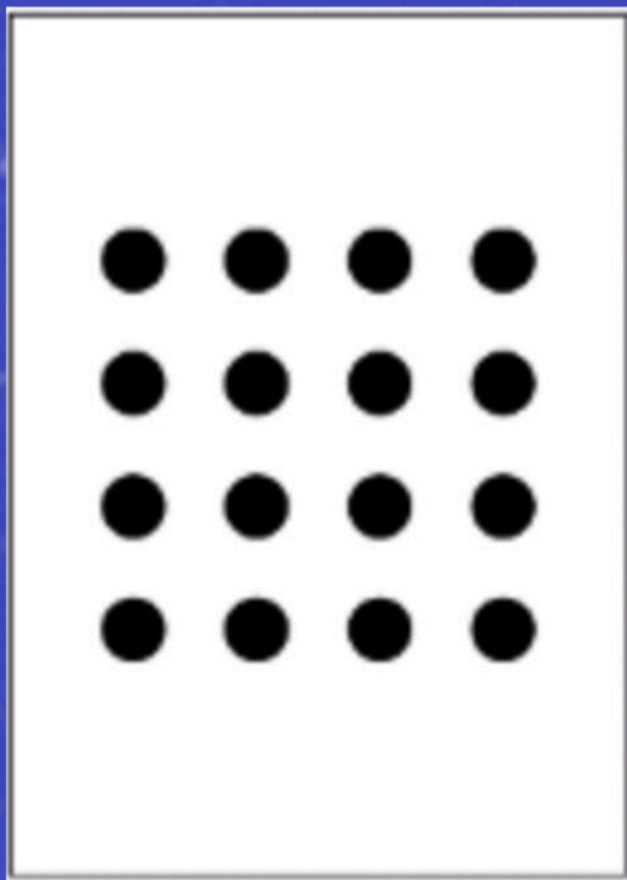


5 = 0101

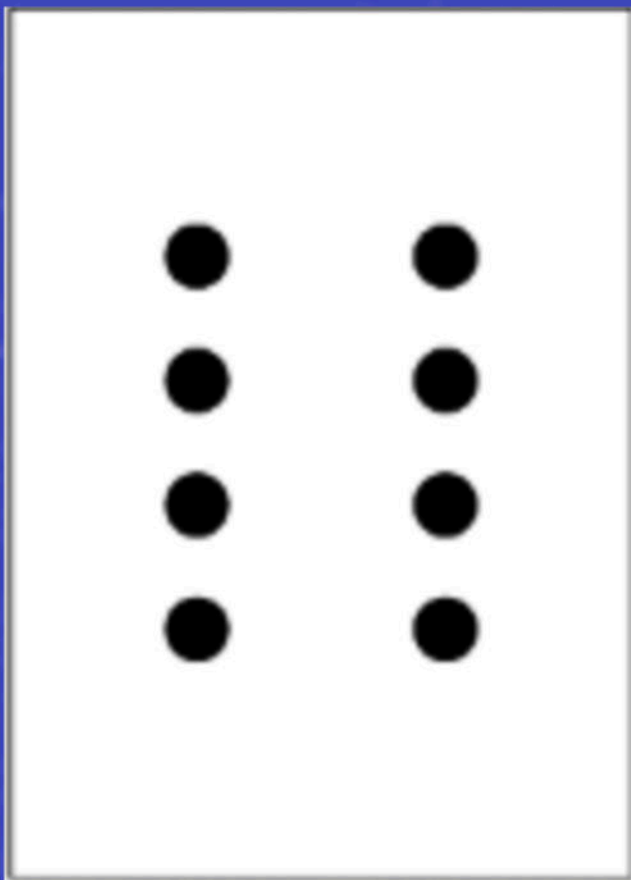
20 = 1010

26 = 11010

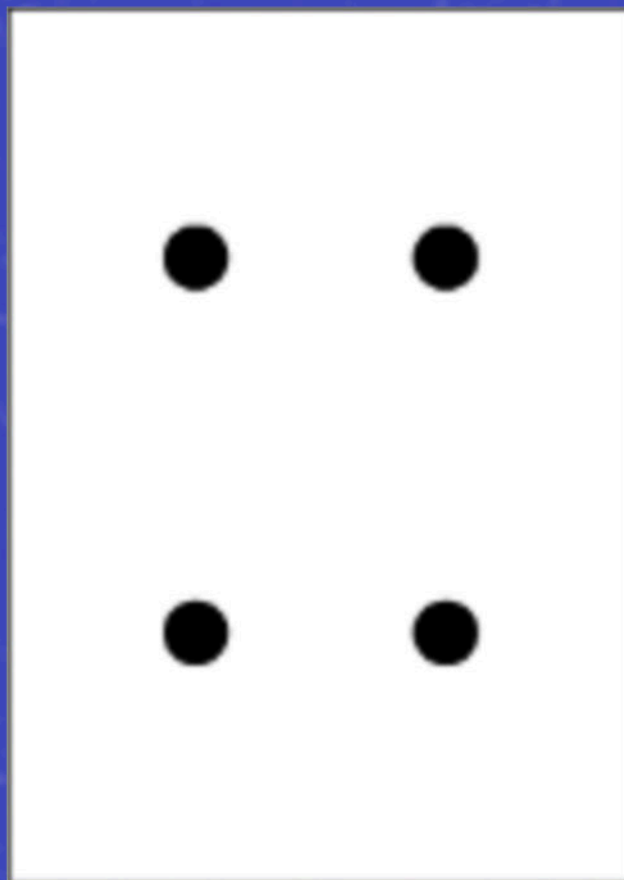
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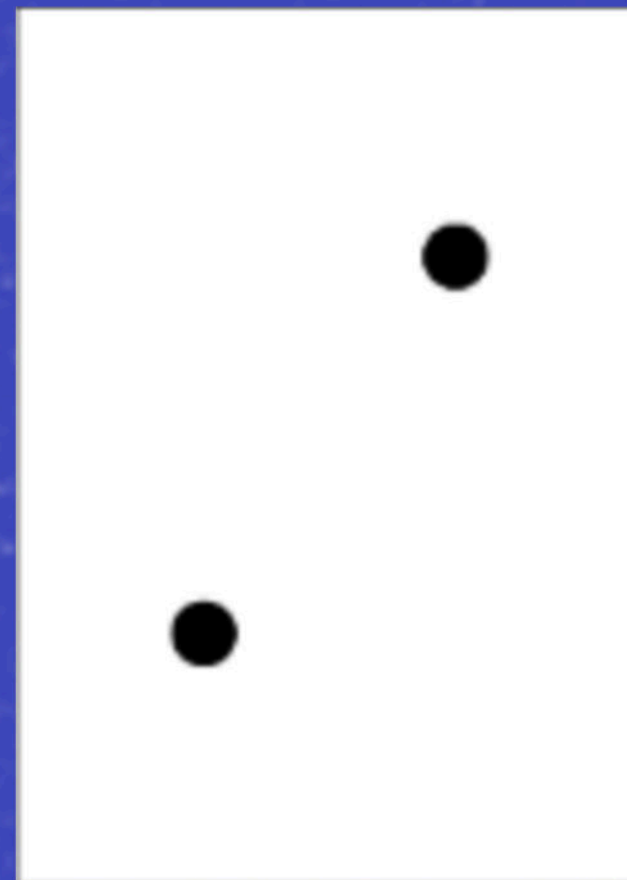
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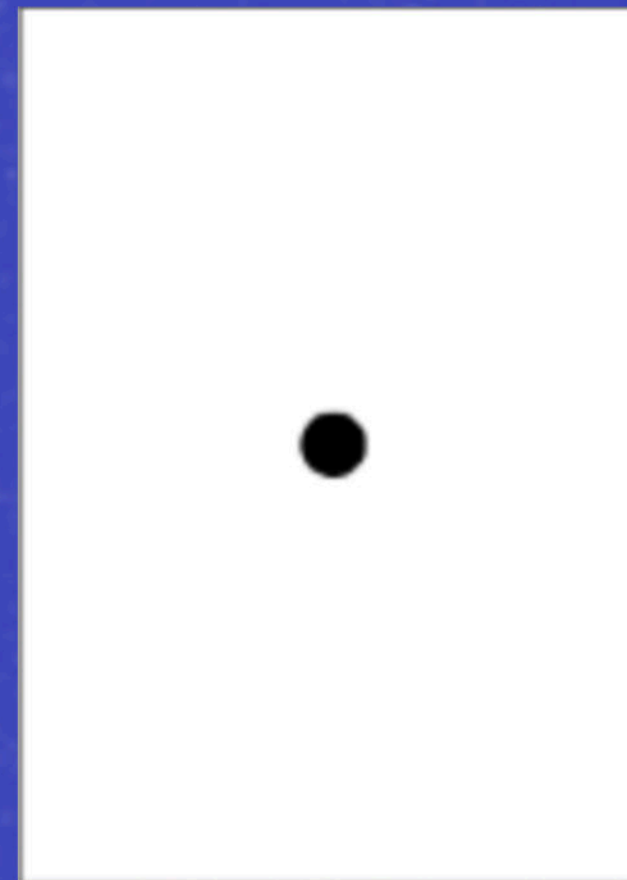
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2



1



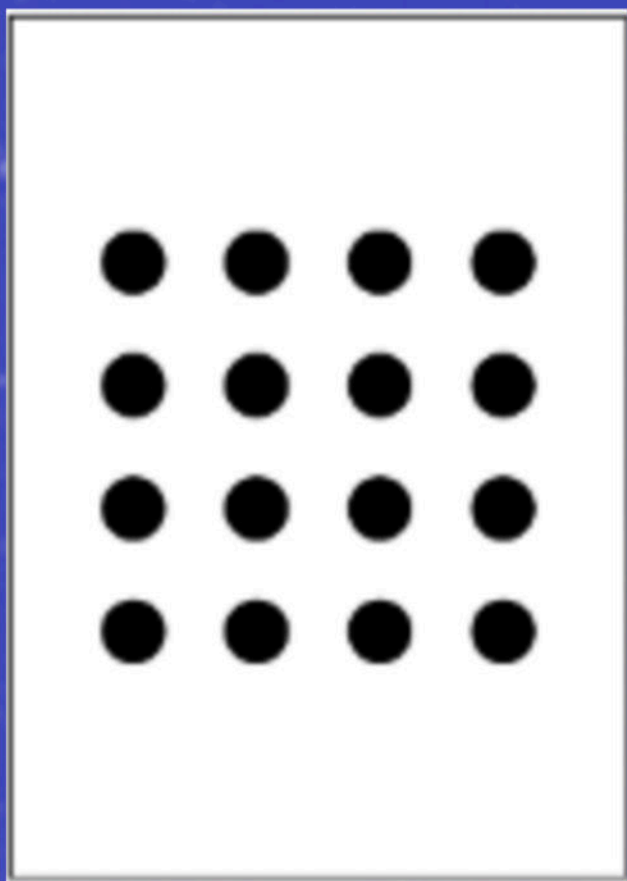
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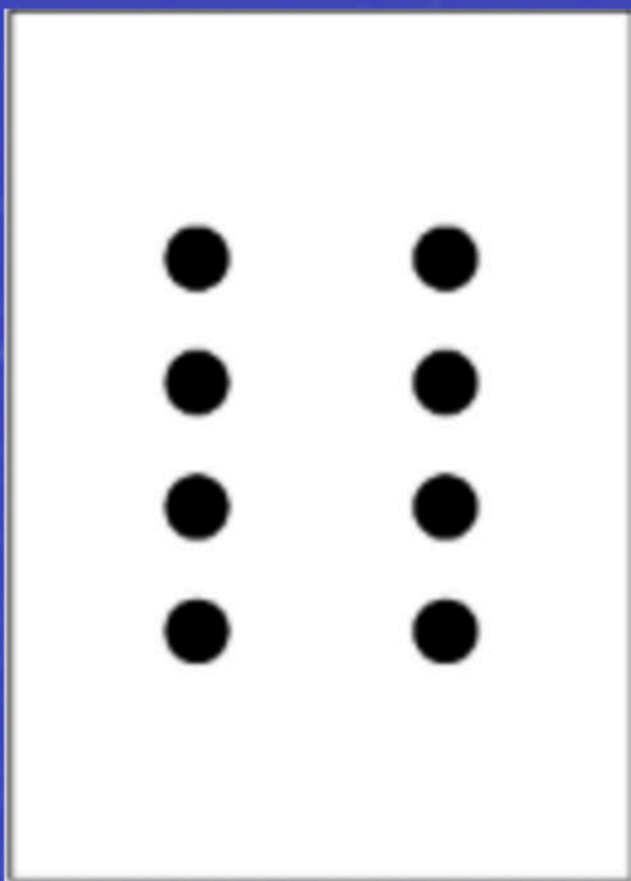
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26 = 11010

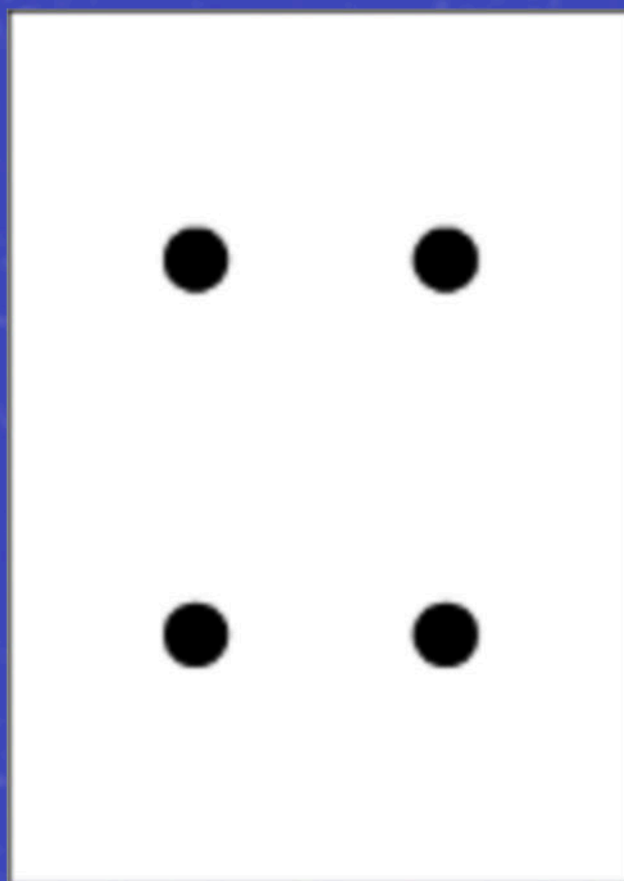
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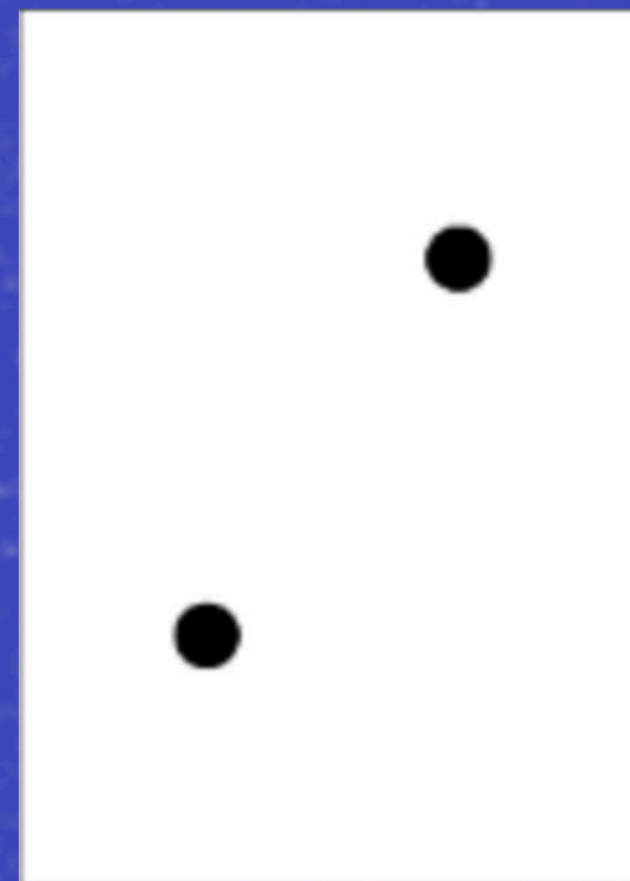
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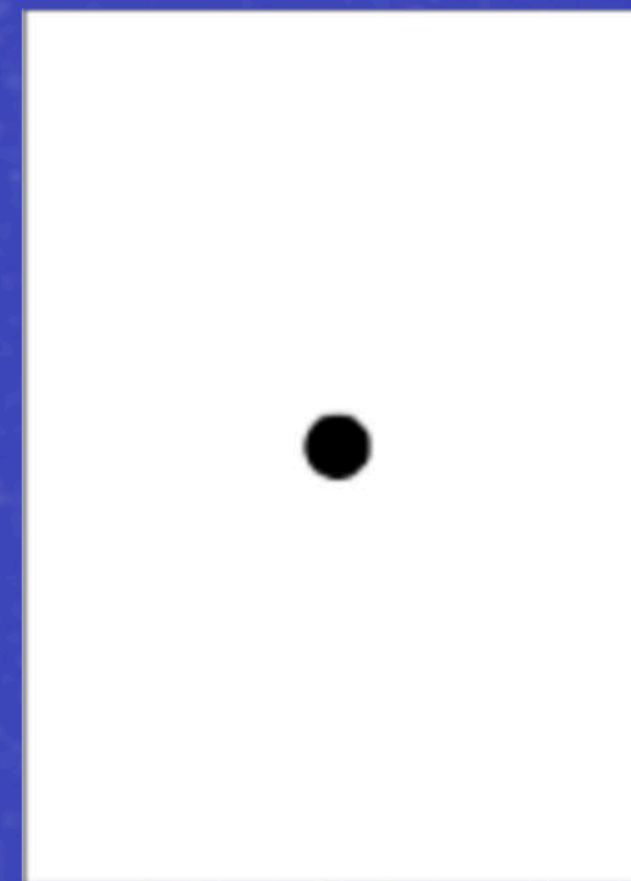
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2



1



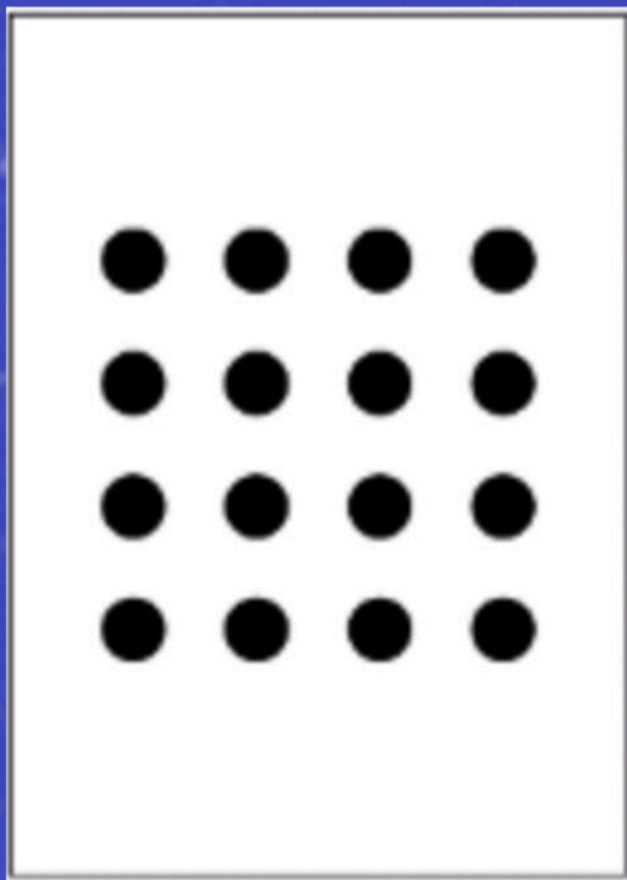
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11 = 01011

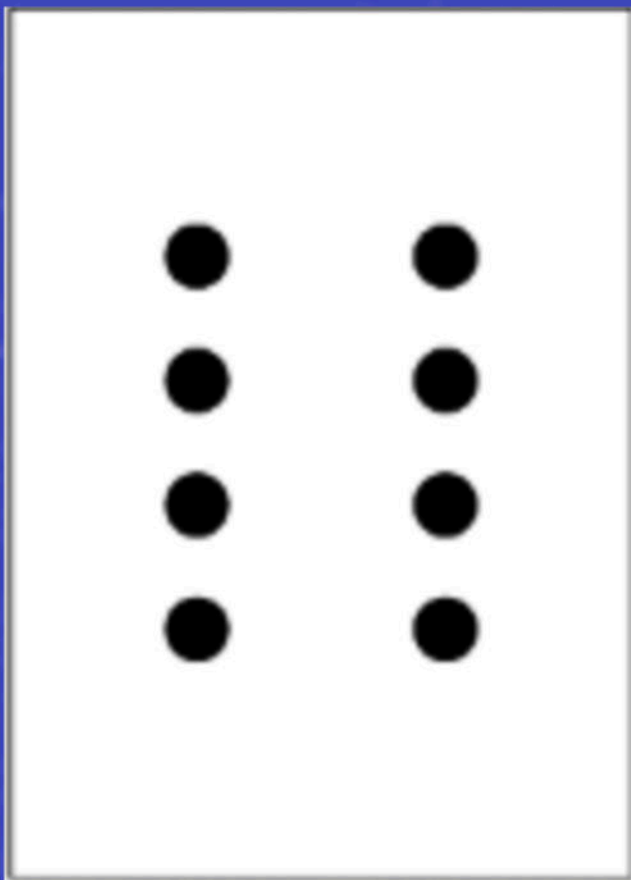
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26 = 11010

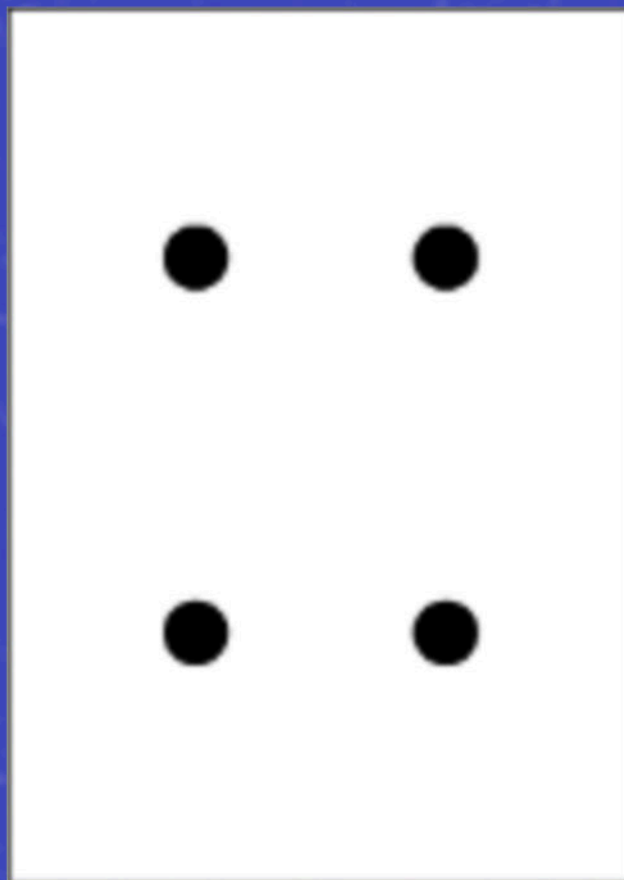
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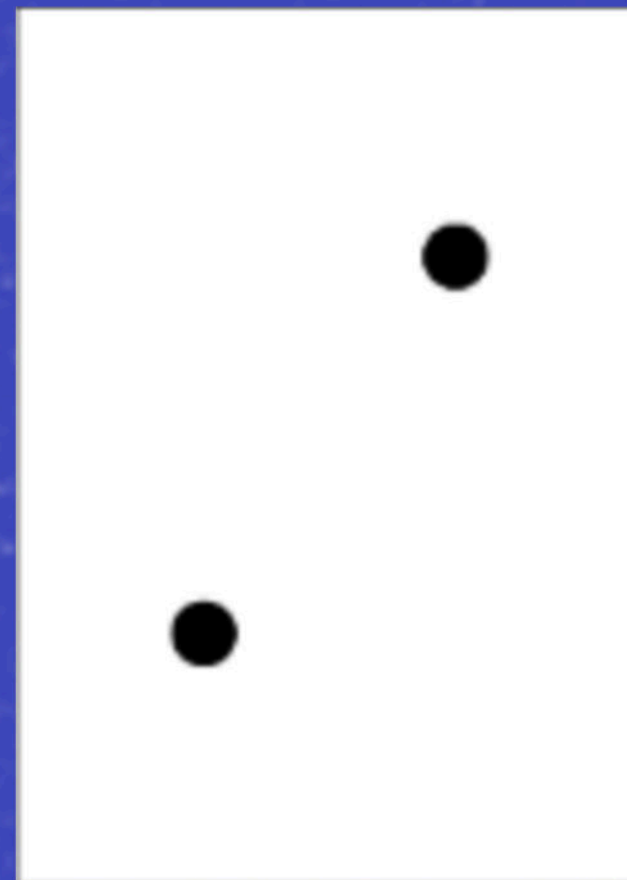
8



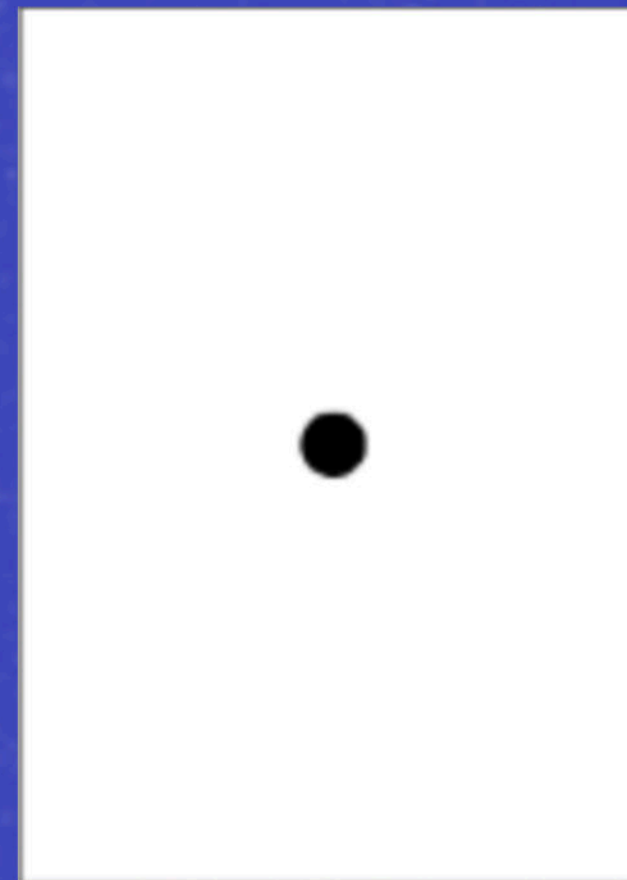
4



2



1



5 = 0101

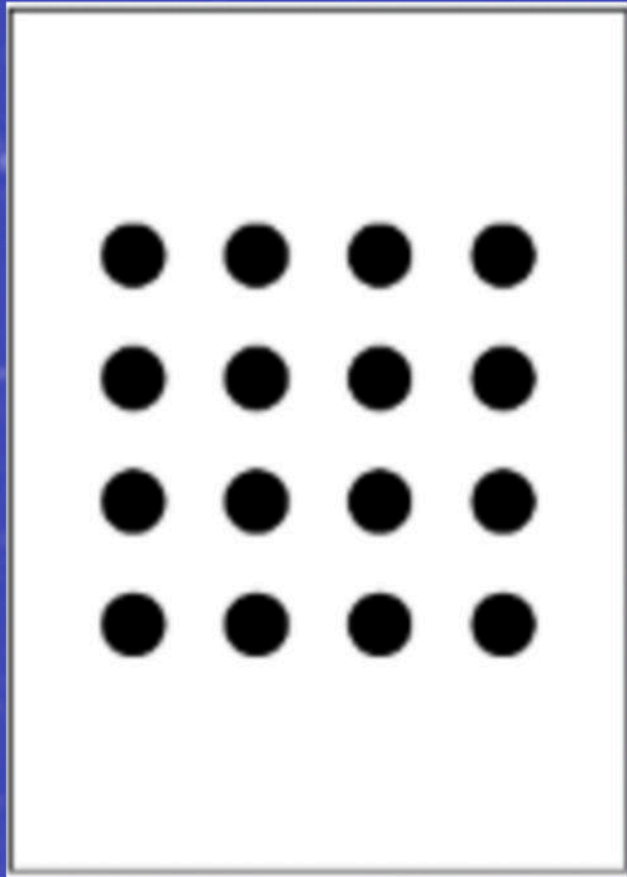
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26 = 11010

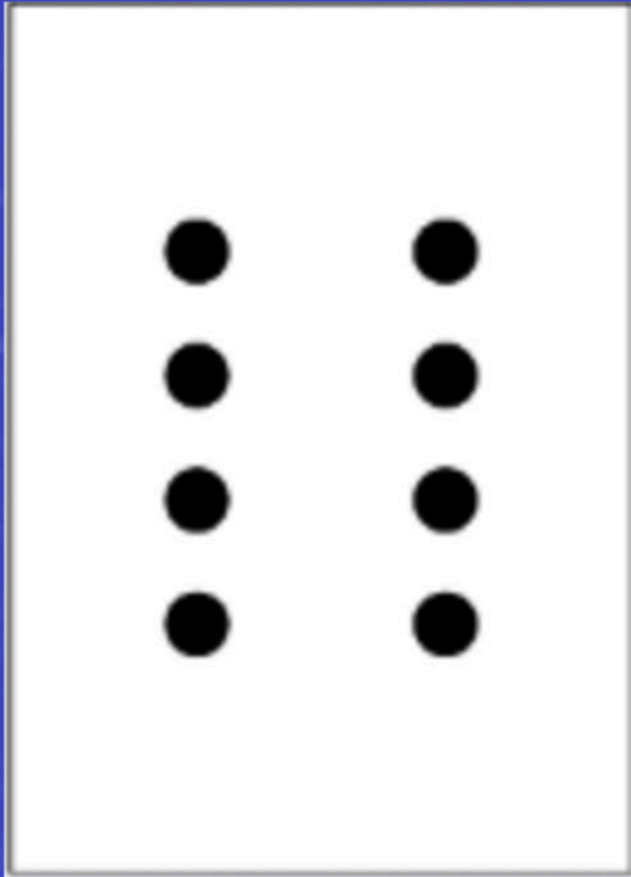
11 = 01011

14 =

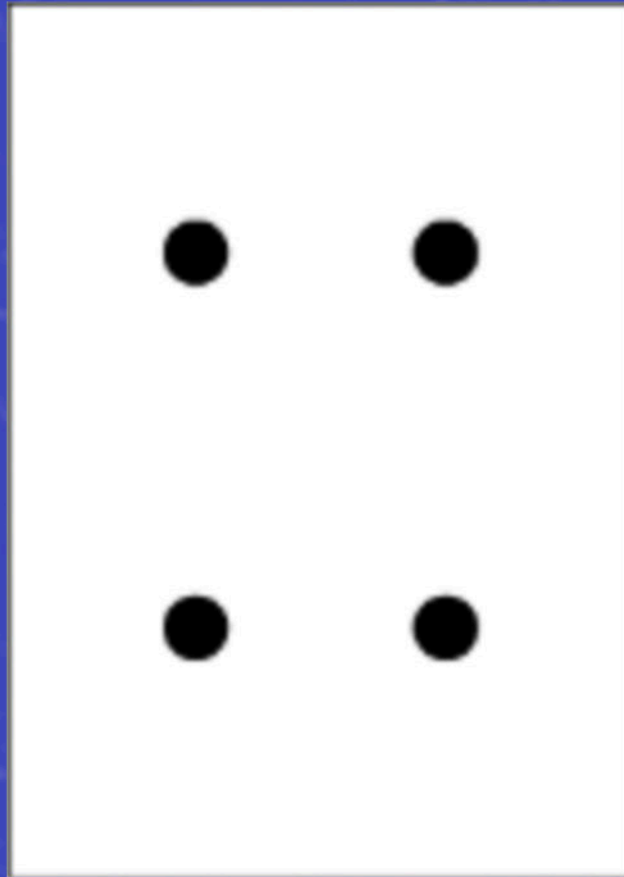
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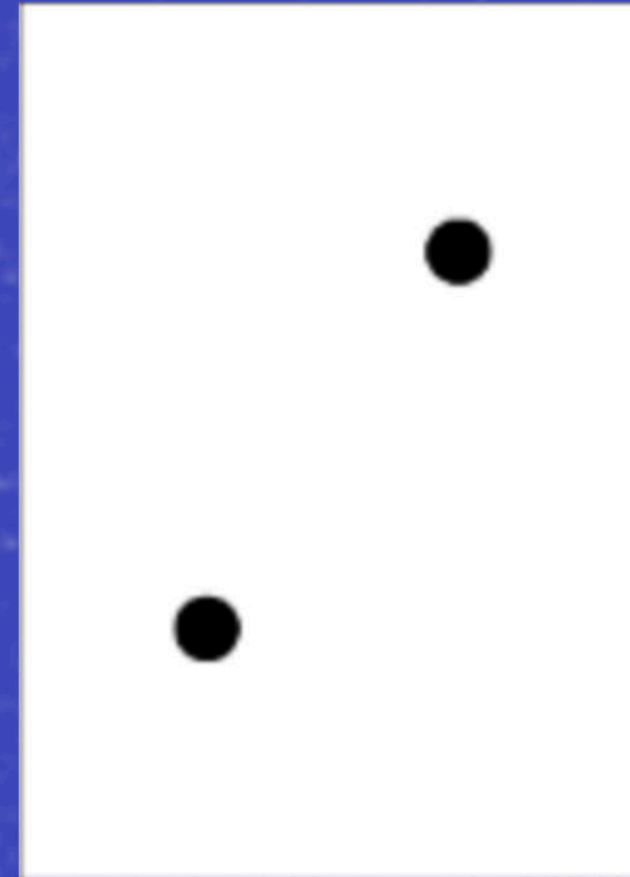
8



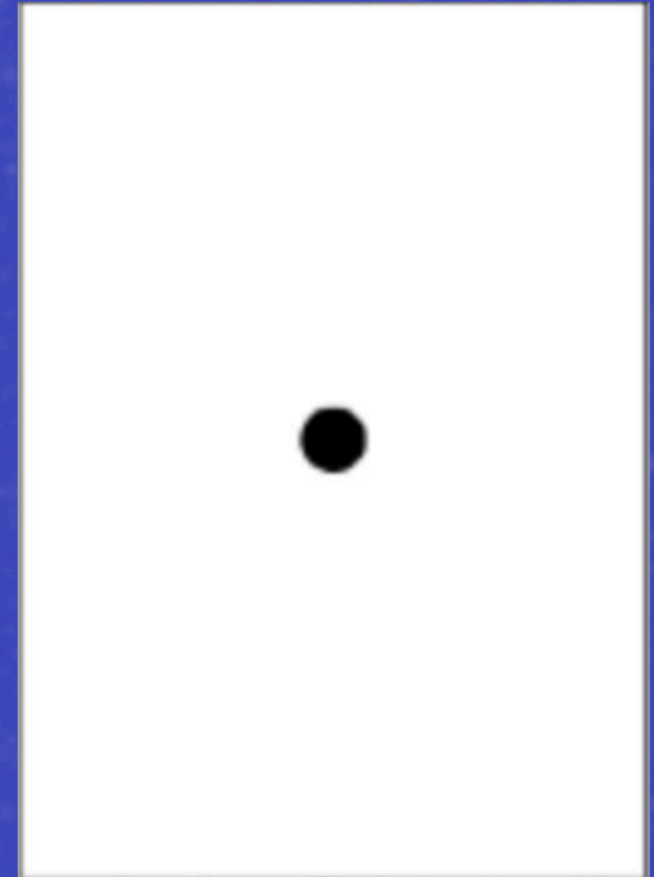
4



2



1



5 = 0101

11 = 01011

20 = 10100

14 = 01110

26 = 11010

Respostas

Converte de decimal para binário:

$$110 = \underline{1101110}$$

$$85 = \underline{1010101}$$

$$30 = \underline{011110}$$

$$14 = \underline{01110}$$

Converte de binário para decimal:

$$101101 = \underline{45}$$

$$01111 = \underline{15}$$

$$01001 = \underline{9}$$

$$0110 = \underline{6}$$

	= 1
	= 10
	= 21
	= 4
	= 5
	= 0
	= 5
	= 19
	= 20
	= 15
	= 21
	= 0
	= 16
	= 18
	= 5
	= 19
	= 15

Obrigado pela atenção!!